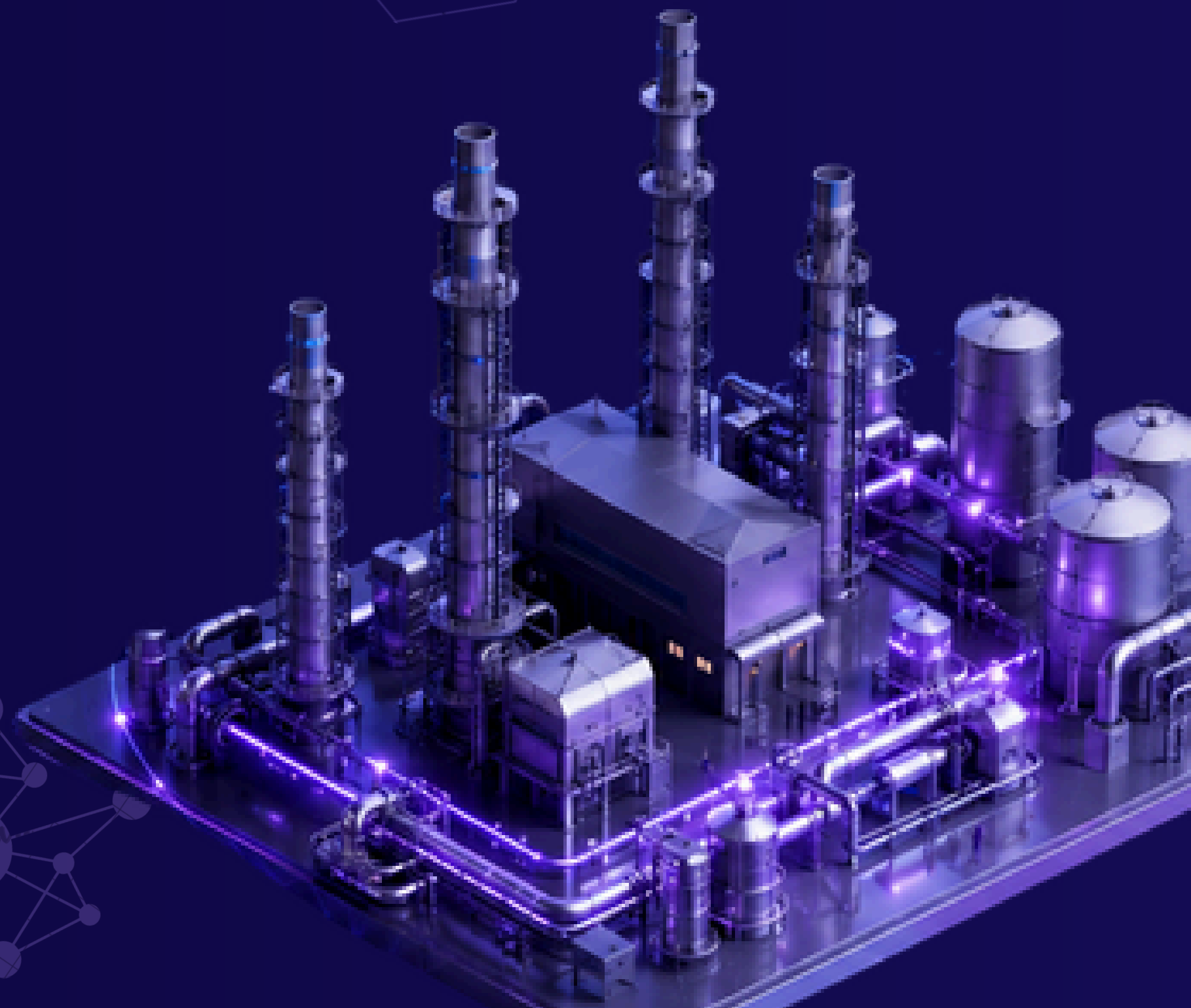




PlantForge

Forging the Unified Asset and Operations Brain





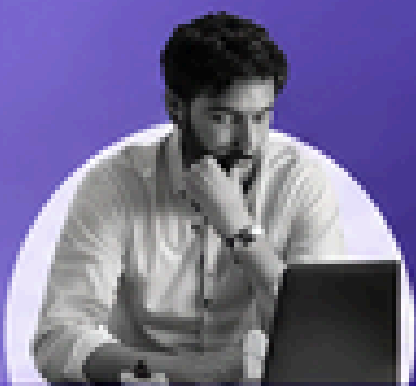
Srinjoy Das
IIT Kharagpur
Computer Science and Engineering



Shamik Bhattacharjee
IIT Kharagpur
Chemical and Financial Engineering



USER PERSONAS AND PRODUCT MARKET FIT



Process Engineer Ben Adams

"How do I pinpoint the root cause of recurring trips without losing 30% of my week searching fragmented P&IDs?"

Core Needs



Topological Tracing



MoC Intelligence



Root Cause Analysis



Verified Context

Pain Points



Slow Research



Black-box Models



Fragmented Data



Lost in the Middle

AI Scope



PathRAG Reasoning: Traversing topological graphs, not just vector similarity



Vision-Language Parsing: Extracting geometry and flow from P&IDs



Autonomous Investigation: Agents that trace root causes before engineers arrive



Deterministic Grounding: Strict verification against source documents



Plant Manager Sara Mathews

"Can I get an instant, audit-ready compliance dossier across all 12 documented software systems?"

Core Needs



Macro Visibility



Compliance Assurance



Actionable Reporting



Knowledge Retention

Pain Points



The Downtime Penalty



Information Overload



Compliance Blindspots



The Knowledge Cliff

AI Scope



Automated Dossiers: Generating executive summaries and mitigation plans



Standards Watcher: Monitoring the web for regulatory code updates



Plant-Wide Digests: Live, aggregable queries for asset health



Strategic Copilot: Answering high-level ROI and reliability queries



Plant Operator Munna Bhai

"Can I receive a verified, hands-free isolation procedure while standing in front of the machine?"

Core Needs



Instant Answers



Hands-Free Access



Precise LOTO



Safe Permits

Pain Points



Complex Interfaces



Dangerous Hallucinations



Generic Manuals



Alone Situations

AI Scope



Voice Pipeline: Low-latency conversational AI via Pigeon and WebRTC



Semantic Answer Cache: Instant responses for pre-computed alerts



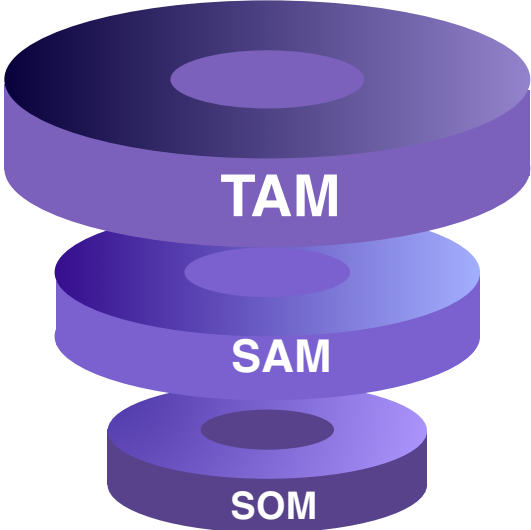
Permit Generation: Drafting hazard mitigations based on topological boundaries



Strict Verification: Blocking any AI-generated claim absent from the manual



MARKET OPPORTUNITY



Total Addressable Market
Global industrial AI market
projected to reach **\$27.8B**
by 2032.

Serviceable Addressable Market
The Knowledge Management &
AI-Augmented Asset Management
market.

Serviceable Obtainable Market
500M+ in spend across heavy
manufacturing and process
industry 4.0.

QUANTIFIED
IMPACT



of time report
AI hallucinations



cannot quantify
AI ROI
internally



fail governance
audits

LIMITATIONS OF LEGACY ENTERPRISE SYSTEMS (ERP, DMS, EAM)



Disconnected
software systems



Loss of Tribal
Knowledge



The "Search Tax"
Burden

ARCHITECTURAL FLAWS OF STANDARD AI & VECTOR RAG



Failure in
Multi-Hop
Relational
Reasoning



The "Lost in
the Middle"
Degradation



Unacceptable
Hallucination
Risks



Purely
Reactive
Design

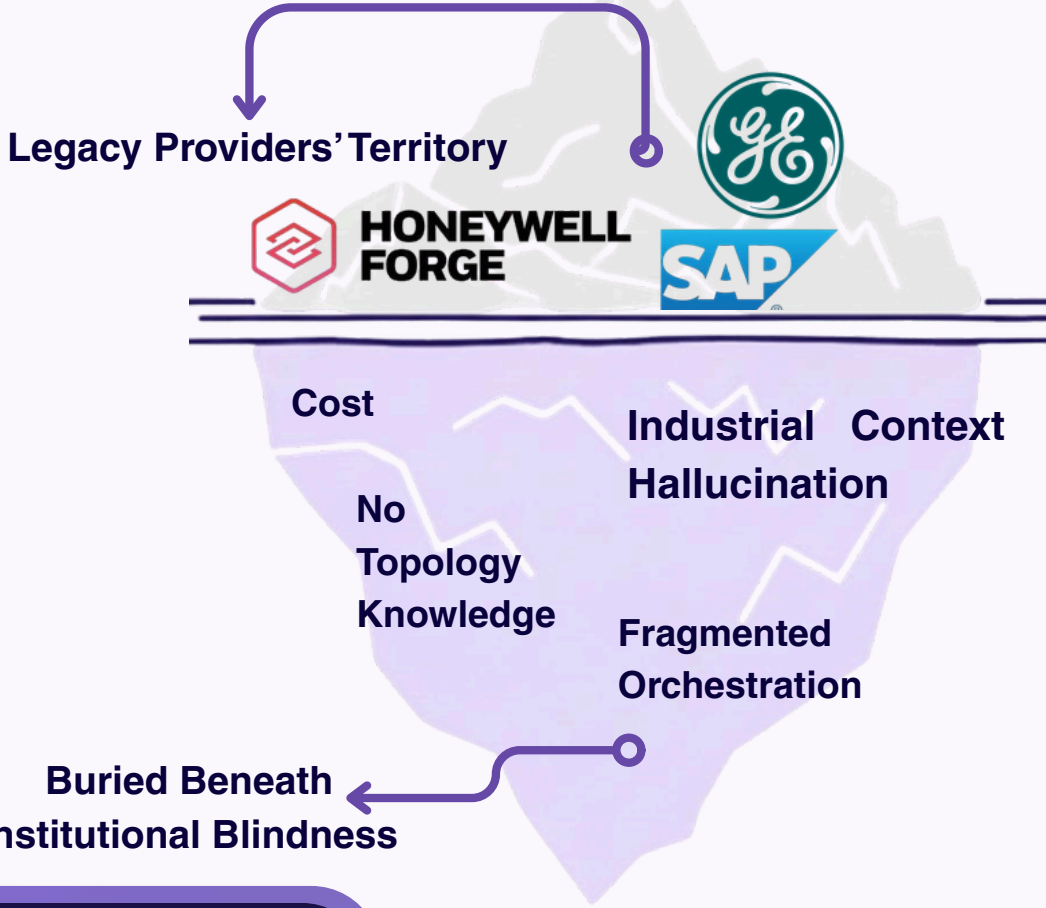
EXTRACTION & ENTITY RESOLUTION BOTTLENECKS



Loss of Visual &
Spatial Layouts



Fragmented
Data Islands



25% of the experienced workforce
is retiring within a decade.



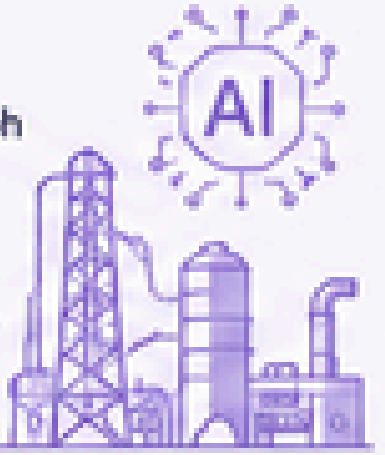
Engineers waste 35% of their time
(1.7 day/week) searching.



Siloed systems span 7 to 12+
fragmented systems.

PlantForge maps
physical reality.

By building our knowledge Graph
from first principles, our engine
understands the actual plant
(topology and fluid mechanics)—
not just word associations.





POWERED BY

PathRAG Neo4j Graph Celery SSE Streaming

Built for Industrial Scale.

A meticulously designed platform that understands your plant's topology and grounds every answer in verifiable reality.



Ask anything, cited

Work orders, P&IDs, SOPs, manuals, emails and photos become one queryable brain. Every answer carries its source.

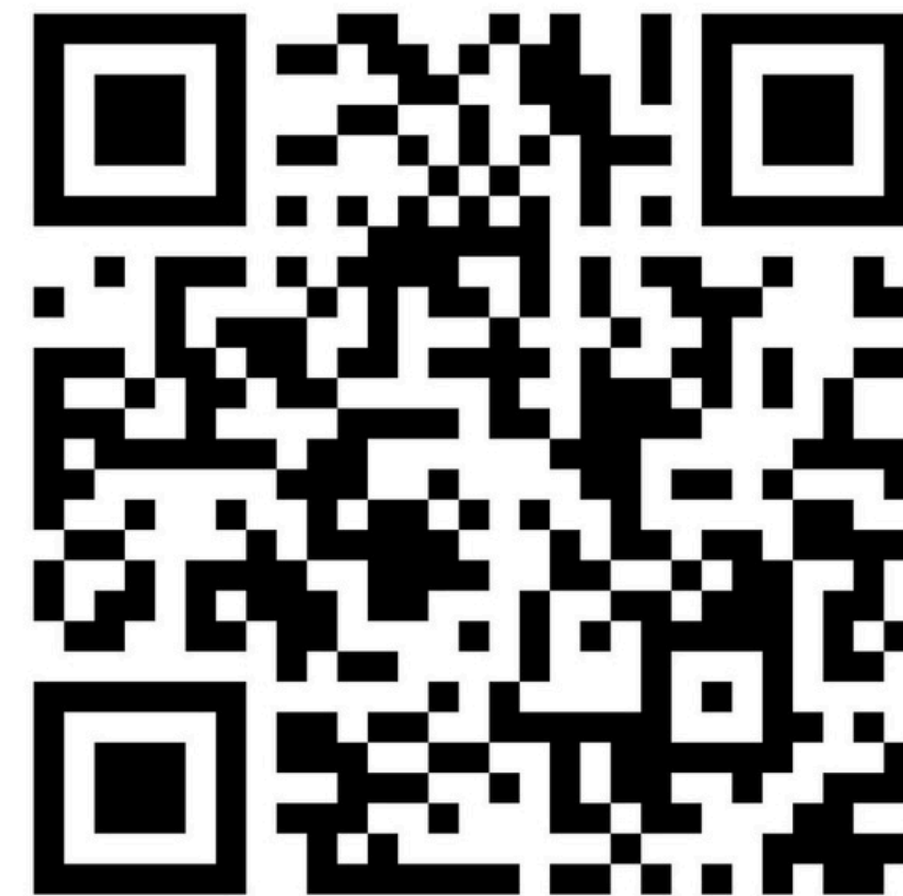


Reasoning you can see

PathRAG traverses the plant's real topology to answer causal questions.



SCAN ME !



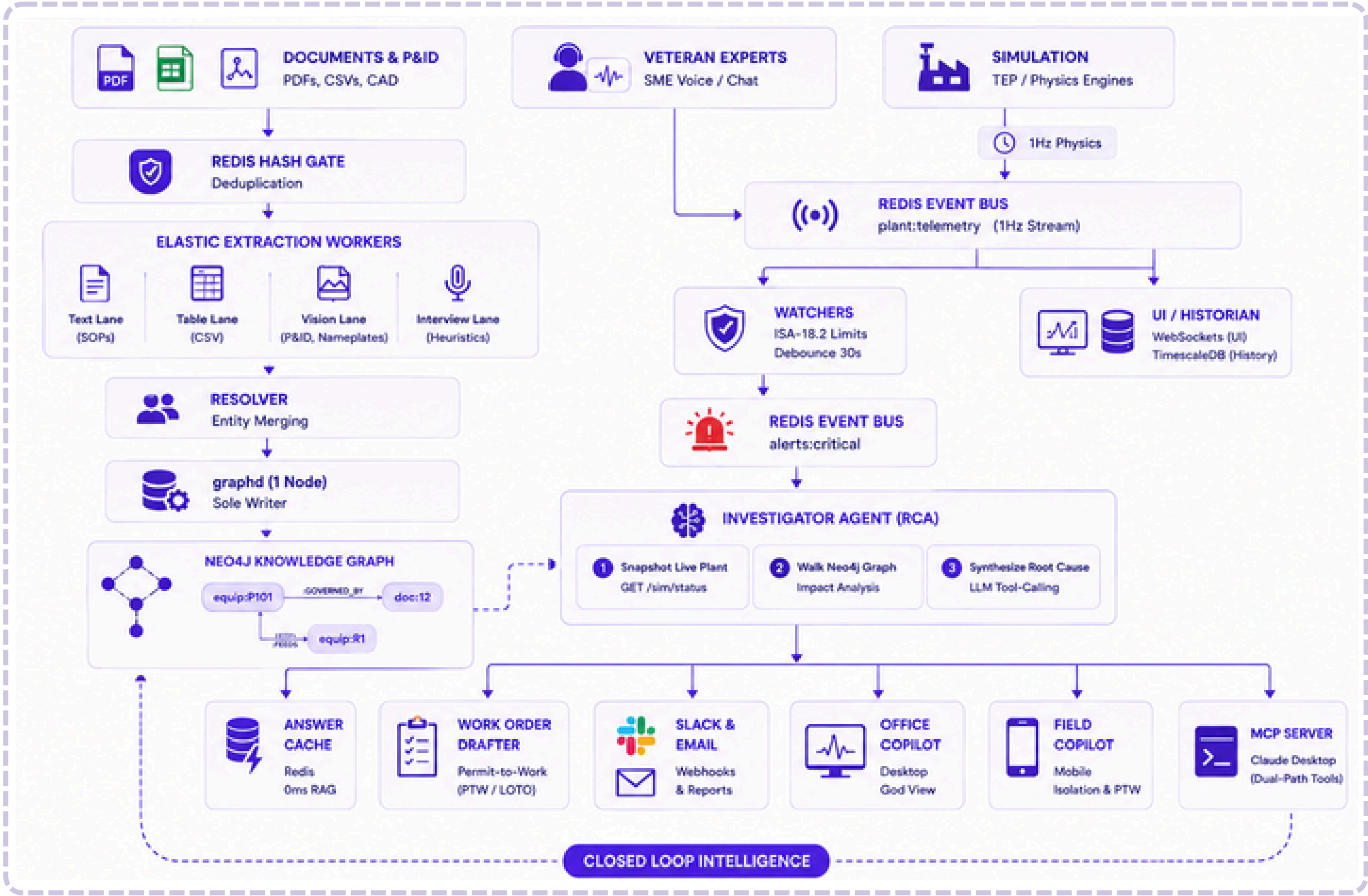
***Step Into the Control Room:
Test-Drive PlantForge Live!***



<https://tinyurl.com/PlantForge>

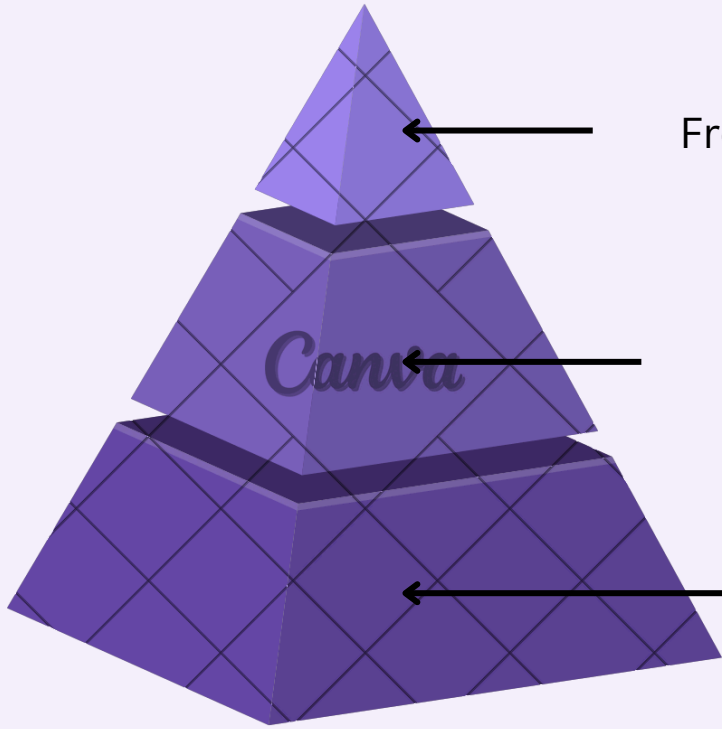


SOLUTION OVERVIEW



PlantForge give the plant its own brain and paces up automation of process industry exponentially with the help of Agentic AI Workflows.

Hierarchical System Design



Frontend and UI/UX

Optimised AI Stack

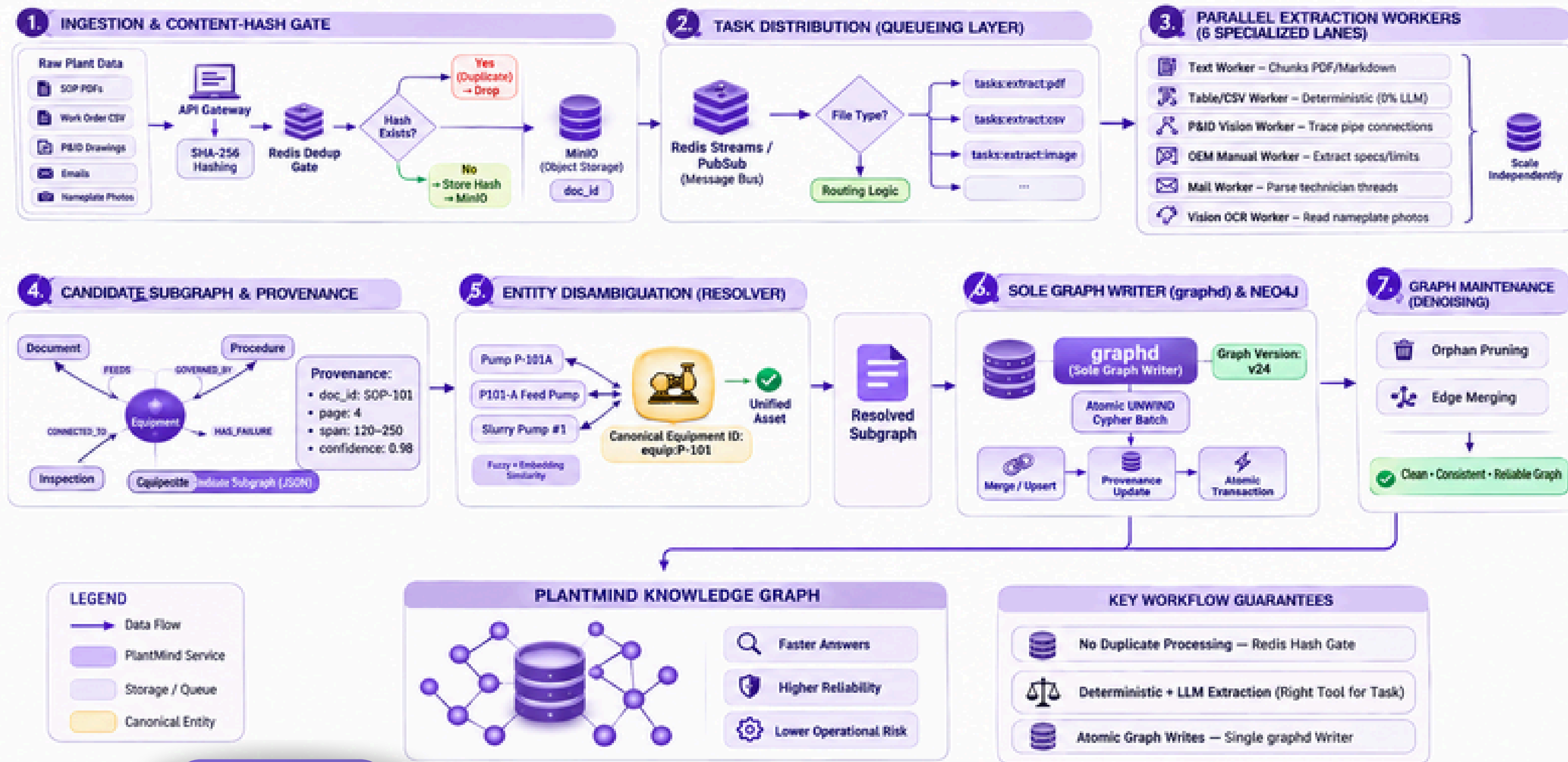
Systems level optimisations and Security

Tech Stack



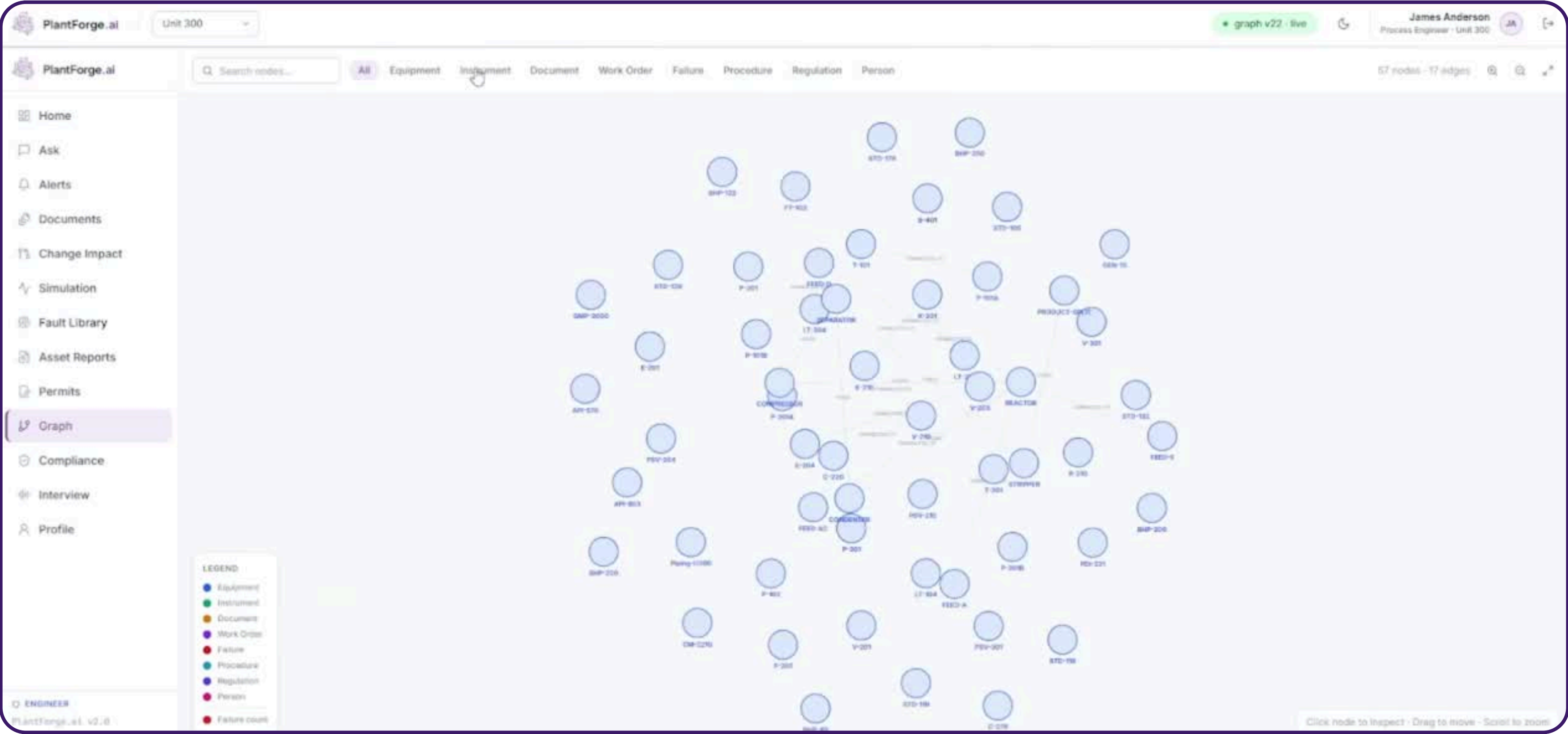


PLANT MEMORY AND KNOWLEDGE GRAPH



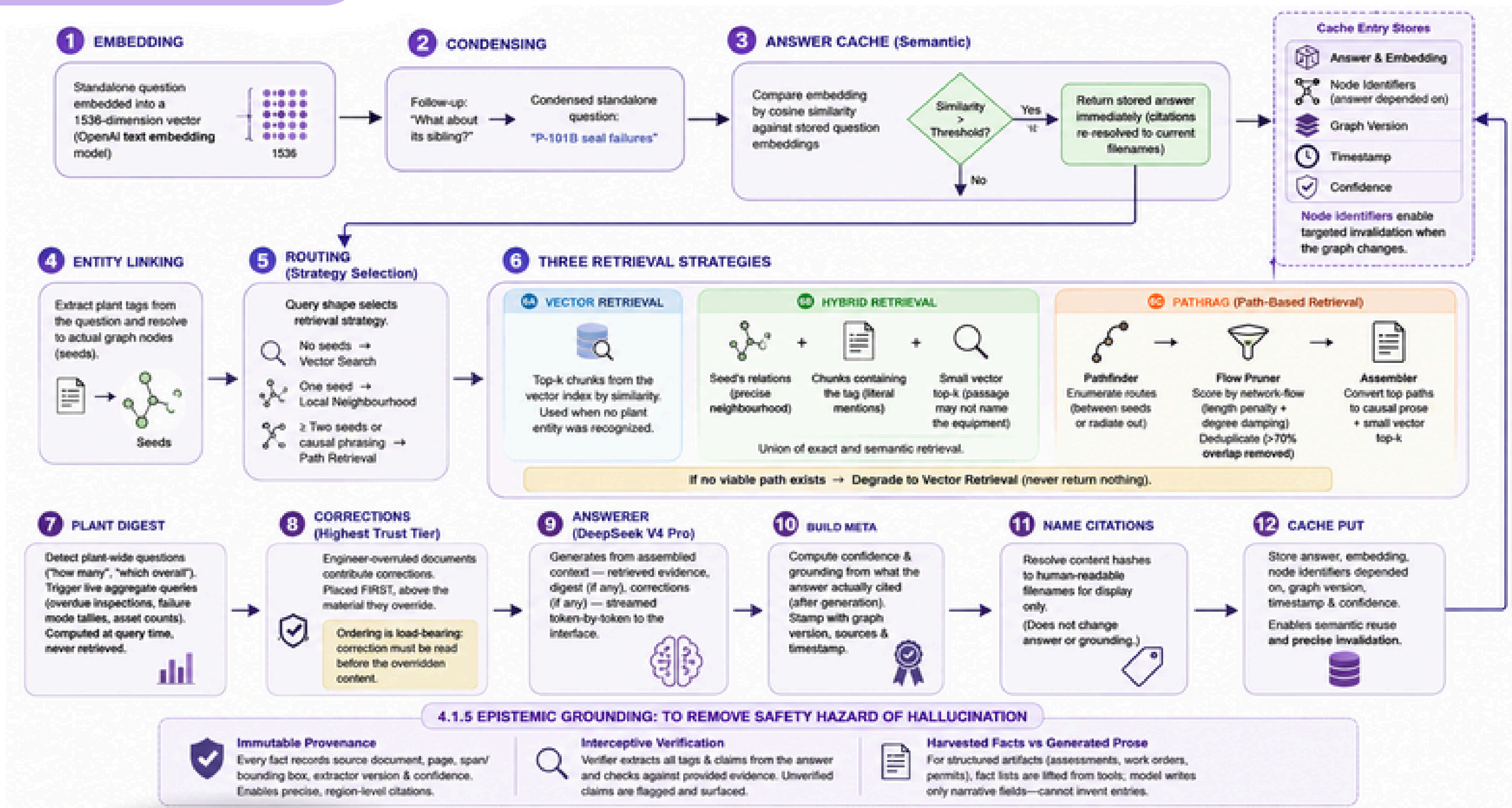


PLANT MEMORY AND KNOWLEDGE GRAPH





RETRIEVAL WITH TOPOLOGICAL PATH RAG PIPELINE





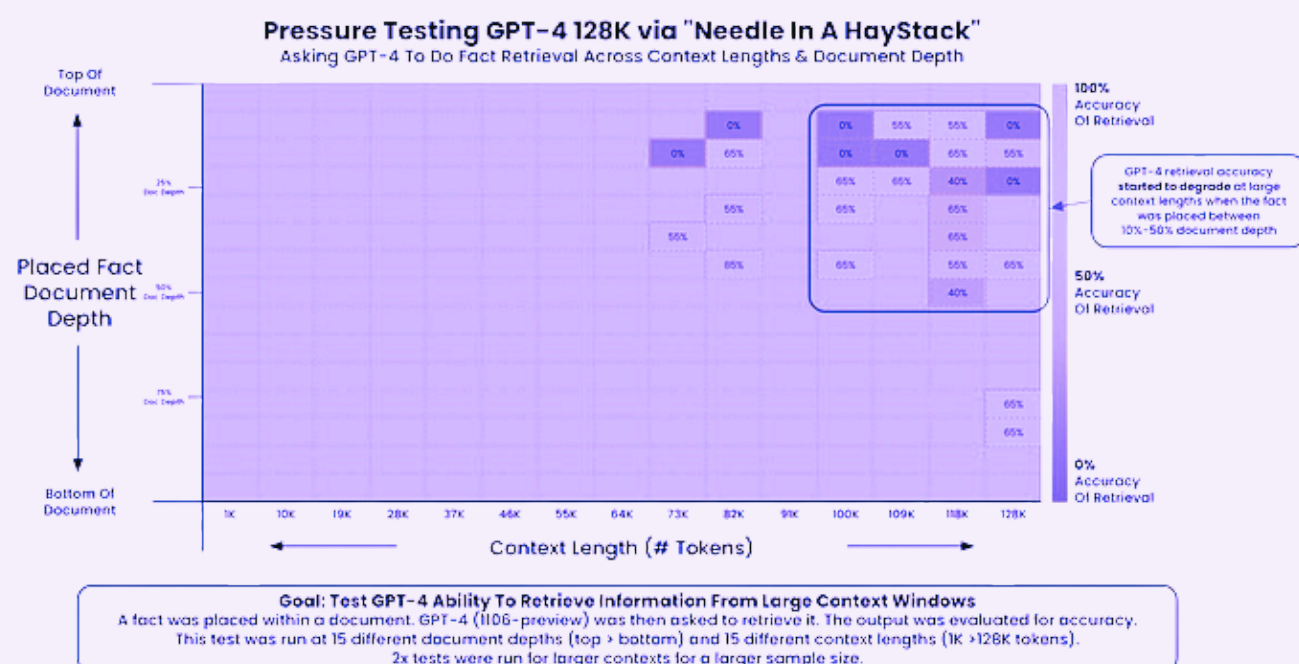
What Do Paths Look Like in PathRAG?

In standard RAG, the data unit is a disconnected, arbitrary text chunk. In PathRAG, the data unit is a **causal topological chain** extracted from the Neo4j knowledge graph.

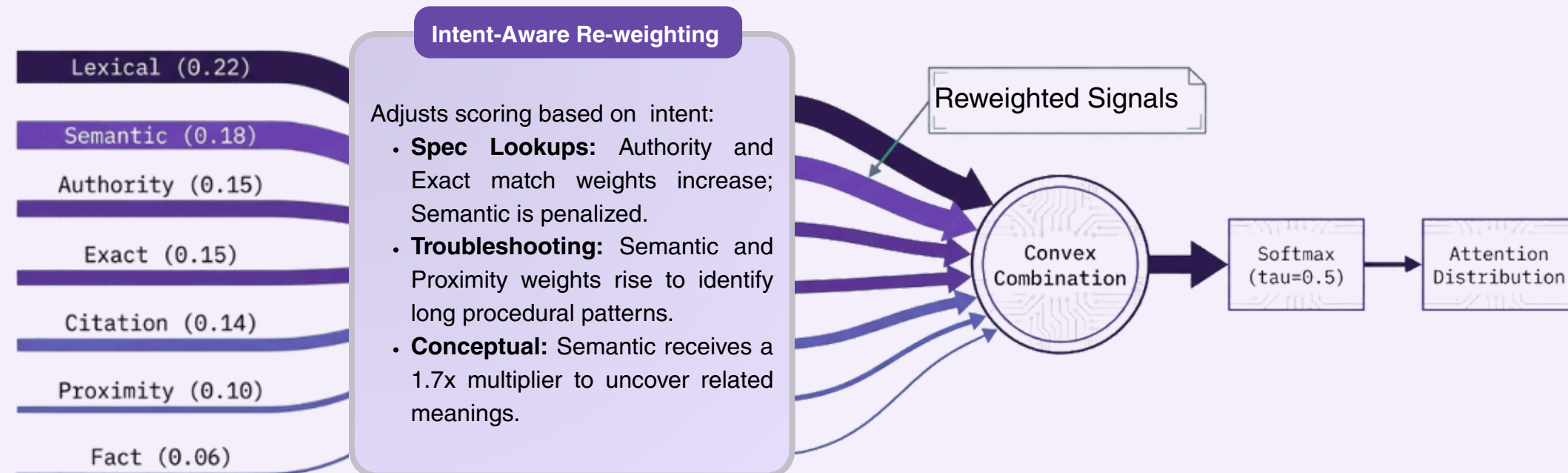
Raw Graph Structure: (Tag: PT-101) -[MEASURES]-> (Equipment: Vessel V-200) -[CONNECTED_TO]-> (Tag: RV-201) -[GOVERNED_BY]-> (Doc: SOP-88)

Serialized Path: "Pressure Transmitter PT-101 measures Pressure Vessel V-200. Vessel V-200 is physically connected via a 4-inch line to Relief Valve RV-201. Relief Valve RV-201 is governed by maintenance procedure SOP-88."

When the system pulls data, it generates dozens of these serialized paths. This collection of paths becomes the **"needle in a haystack"** problem



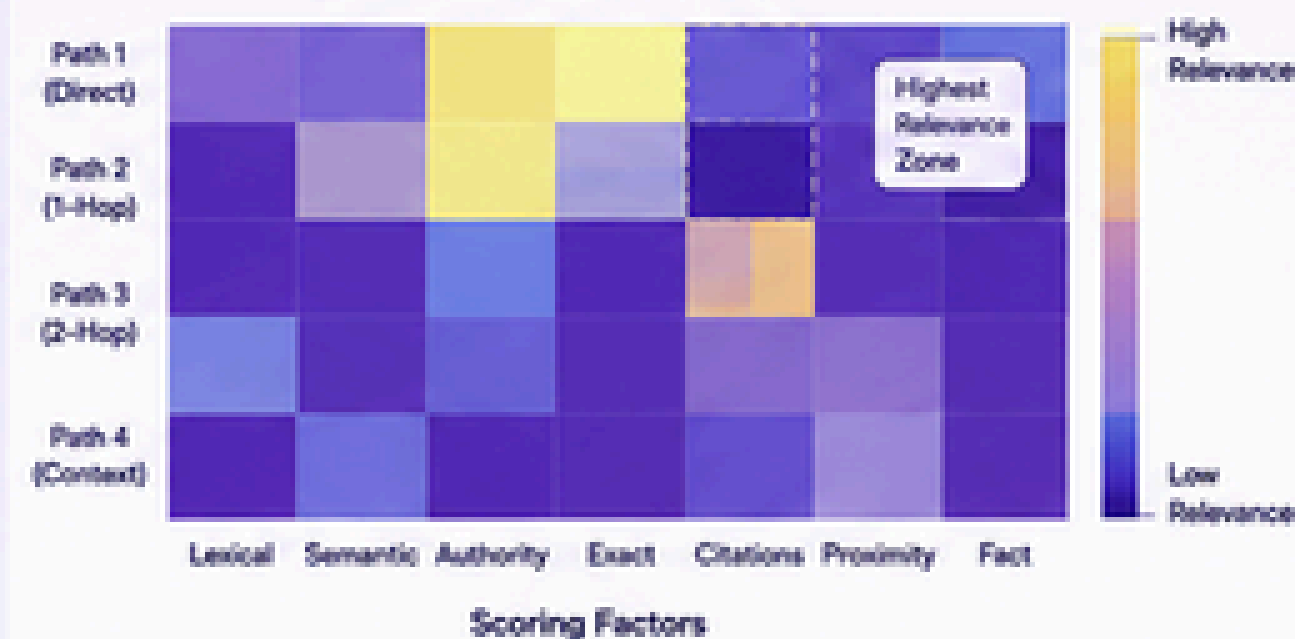
Hybrid Attention Scoring



Spatial context engineering

- The LLM does not see a messy graph database; it receives a rigidly structured prompt.
- The absolute highest-scoring path (e.g., the direct causal link) is **pinned to the very top**. The second highest-scoring path is **pinned to the very bottom**, right next to the user's query following a permutation algorithm

Attention Distribution Heatmap (Combined Graph Rerank)





RAG EVALUATION - WHY WE BEAT OTHERS?

1 ARCHITECTURAL CAPABILITY COMPARISON

Dimension	VectorRAG	GraphRAG	LightRAG	RAPTOR	PlantForge
Retrieval Unit	Passage	Community Summary	Entity + Theme	Summary Tree	Path
Multi-hop Causal Traversal	✗			✗	✓
Path Redundancy Control	N/A	✗	✗	N/A	
Extraction Confidence Weighting	✗	✗	✗	✗	✓
Incremental Update (No Re-index)	✓	✗	✓		✓
Multimodal Ingest (P&ID / Tables)	✗	✗	✗	✗	✓
Hallucination Verification	✗	✗	✗	✗	✓

2 GOLDEN SET PERFORMANCE (CONTROLLED ABLATION)

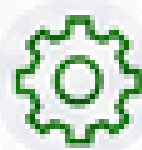
Metric	VectorRAG	GraphRAG	LightRAG	PlantForge
Recall@5 (Multi-hop Queries)	34.2%	59.3%	67.8%	81.1%
Citation Accuracy	58.1%	73.5%	72.3%	86.8%
Context Tokens Per Query	~4,500	~12,800	~10,900	~9,200
Retrieval Latency (p50, ms)	320 ms	1,450 ms	1,620 ms	890 ms

3 "LOST IN THE MIDDLE" TEST (NIAH)



PlantForge maintains >85% accuracy even when the fact is buried in the middle.

4 CUSTOM INDUSTRIAL GOLD STANDARD (TEP / NASA / OEM DATA)



Exact Tag Match Rate

Alphanumeric tag precision

94.7%

(Higher is better)



Topological Blast Radius Recall

Upstream / downstream coverage

92.3%

(Higher is better)



Interceptive Grounding Trigger Rate

Correct refusal when information missing

100%

(Target: 100%)

LEGEND



YES / Supported



PARTIAL / Limited



NO / Not Supported



Best Performance



Strong Performance



SIMULATION AND REAL TIME PROCESS INTELLIGENCE

1. WHY TEP?



Industry benchmark
(Downs & Vogel, 1993)



Highly non-linear,
unstable process



Known ground truth
for AI validation

Complex units.
Interdependent.
Perfect for testing AI.

2. WHAT IS SIMULATED?

41 MEASURED VARIABLES (Sensors)



- Reactor Temp
- Reactor Pressure
- Separator Level
- Stripper Level
- Compositions (A-H)
- ... and more

12 MANIPULATED VARIABLES (Actuators)



- A-Feed Flow Valve
- Reactor Cooling Flow
- Purge Valve
- ... and more

20 IDVs (DISTURBANCES) (Inject Chaos)



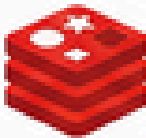
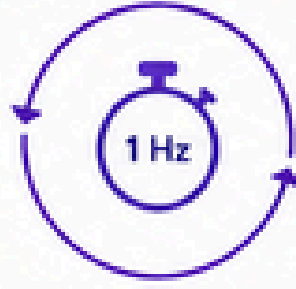
- IDV-1: Feed Ratio Step
- IDV-4: Cooling Temp Spike
- IDV-6: A-Feed Loss (Drift)
- IDV-14: Cooling Valve Sticks
- ... and more



Physics engine solves differential equations ($\Delta t = 1$ sec)
Nominal steady state → Disturbances → Non-linear responses

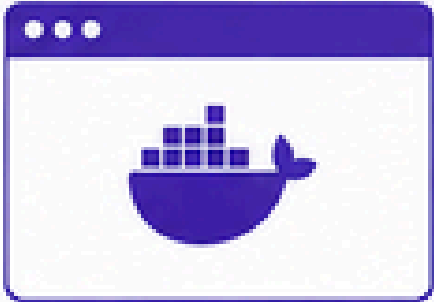
3. HOW IT OPERATES

Continuous 1 Hz Loop (Real Time)



Publish Telemetry
to Redis Pub/Sub
plant-telemetry

TEP SIMULATOR Docker Container



- ✓ Headless & 24/7
- ✓ Deterministic Physics
- ✓ Independent of UI
- ✓ Scales Horizontally

4. DYNAMIC FAULT INJECTION Simulating Real-World Chaos

20 Instructional Disturbance Variables (IDVs)
= Scientifically modeled changes to physics.



IDV-1 A/C Feed Ratio Step Change
Sudden drop in A/C feed ratio.



IDV-4 Cooling Water Temp Spike
Reduces cooling efficiency plant-wide.



IDV-6 A-Feed Loss (Slow Burn)
Creeping degradation, hard to detect.



IDV-14 Sticky Cooling Valve
Valve jams in position despite command.

... and 14 more IDVs

Execution Workflow

1. POST /sim/fault (Inject IDV)
2. Physics matrices updated instantly
3. Thermodynamic ripple effect propagates
4. Telemetry reflects real-world impact

ONE TELEMTRY STREAM. FOUR POWERFUL CONNECTIONS.



Redis Pub/Sub

1

DETERMINISTIC WATCHER (tep_watcher.py)



- Subscribes to telemetry
- Compares to ISA-18.2 limits
- Debounce (e.g., 30 sec)
- Sends Alert to **alertcritical**

2

LIVE UI (GATEWAY WEBSOCKETS)



- Streams 1 Hz data via WebSocket
- Real-time charts
- Dynamic P&ID
- Live plant visibility

3

CONTEXT SNAPSHOT API (GET /sim/status)



- Investigator Agent calls on alarm
- Returns live snapshot + active IDVs
- Grounds RCA in facts

4

Q&A RETRIEVAL ENGINE (Copilot Interrogation)



- Pulls live snapshot
- Injects into LLM context as [doc:tep-sim]
- Precise, real-time answers



RESULT: Real-time physics + deterministic detection + intelligent reasoning = Autonomous operations you can trust.



PlantForge.ai

Unit 300

graph v22 - live

James Anderson
Process Engineer - Unit 300

PlantForge.ai

Home

Ask

Alerts

Documents

Change Impact

Simulation

Fault Library

Asset Reports

Permits

Graph

Compliance

Interview

Profile

Tennessee Eastman Process sim online

Downs & Vogel (1993) benchmark — 8 species · 4 reactions · 6 unit areas · 21 IDV faults

Configure Limits

Reset Views

TENNESSEE EASTMAN PROCESS Online Sim: nominal tick #472

Start

Stop

Reset

IDV Fault Injection

3 active

IDV-14: Reactor coolant valve stuck

Inject

Clear All

IDV-4

IDV-6

IDV-14

IDV-4: Reactor coolant inlet T — step UP

IDV-6: A feed loss (stream 1 → 0)

IDV-14: Reactor coolant valve stuck

REACTOR

CONDENSER

SEPARATOR

STRIPPER

COMPRESSOR

PRODUCT-SPLIT

TENNESSEE EASTMAN PROCESS

Process Flow Diagram

P&ID

Time-Series

Phase Portrait

Live Readings

Alerts

Diagnose

Historian

Live Diagnosis

matched against the simulator's fault library

Investigate with AI

REACTOR.xF

39m ago

No known fault mode matches this signature — an unseen disturbance, or the library has no case for it yet.

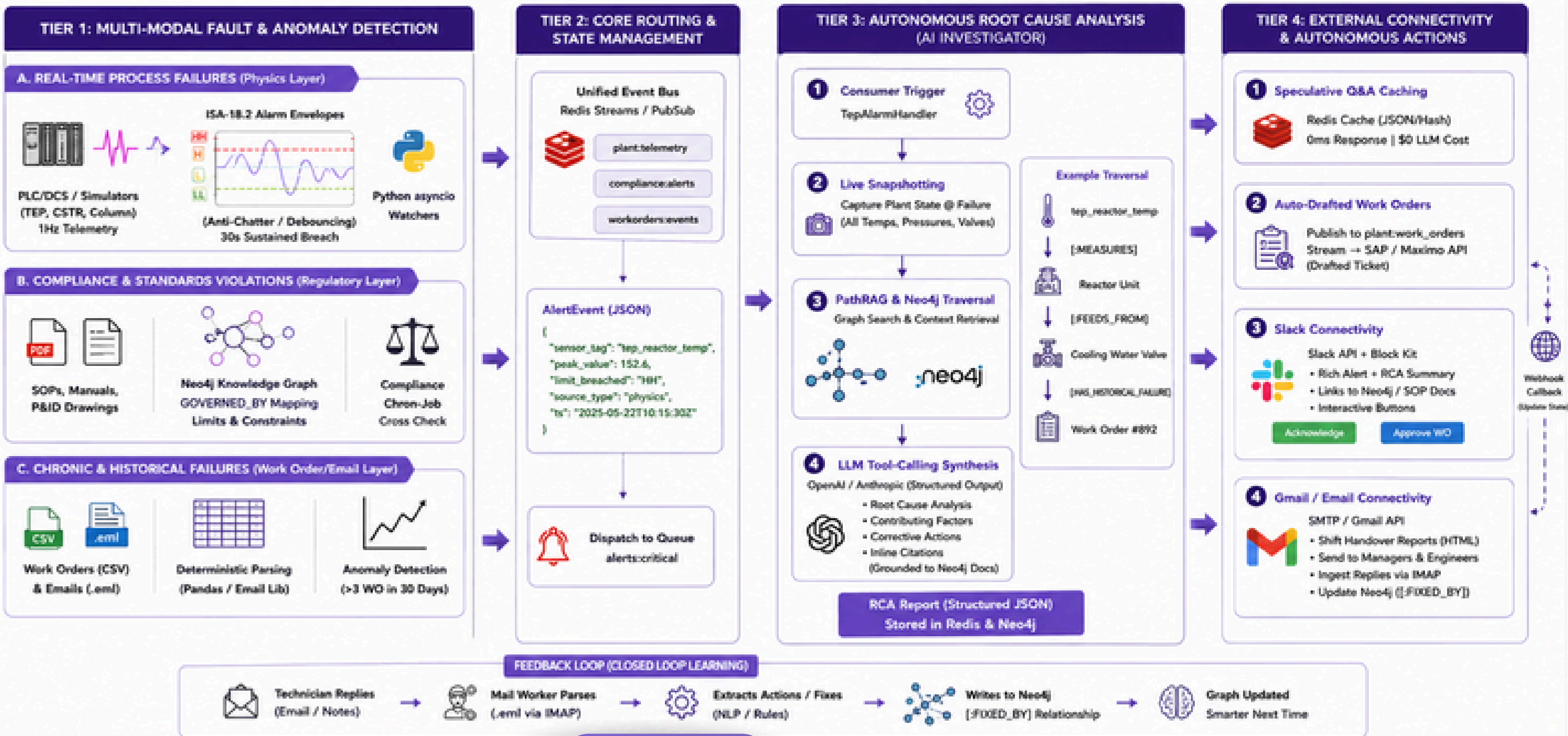
Investigate with AI

ENGINEER

PlantForge.ai v2.8



FAULT DETECTION AND RCA SPECIFIC AMBIENT AGENTS





FAULT DETECTION AND RCA SPECIFIC AMBIENT AGENTS

PlantForge.ai

Unit 300

graph v22 - live

James Anderson
Process Engineer - Unit 300

PlantForge.ai

Home

Ask

Alerts

Documents

Change Impact

Simulation

Fault Library

Asset Reports

Permits

Graph

Compliance

Interview

Profile

Alerts live

Agents watch every failure and compliance event in real time

All Failures Compliance Standards

Alerts 67 Work Orders

67 total 25 failures 12 compliance 30 standards

8-481

Obligation: Hydrostatic test under IBR.

Asset: "8-481"

Due Date: 2026-07-28 (Past Due)

Last Done: 2025-07-28

Immediate maintenance action required to maintain statutory operating compliance.

Statutory inspection required — view compliance position & schedule inspection

SOURCES

IBR (p.72)

Statutory Inspection Overdue: PSV-301 (IBR/OISD-STD-132) critical

PSV-301

Obligation: Pop test under IBR/OISD-STD-132.

Asset: "PSV-301"

Due Date: 2026-08-11 (Past Due)

Last Done: 2025-08-11

Immediate maintenance action required to maintain statutory operating compliance.

Statutory inspection required — view compliance position & schedule inspection

SOURCES

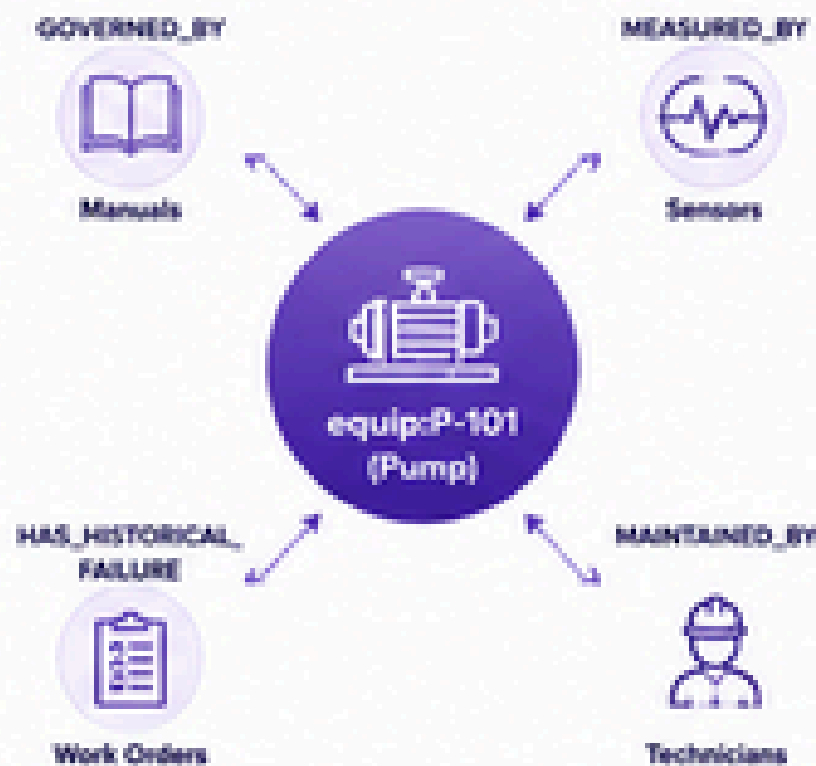
IBR/OISD-STD-132 (p.8)

ENGINEER
PlantForge.ai v2.0



1 ASSET-SPECIFIC INTELLIGENCE

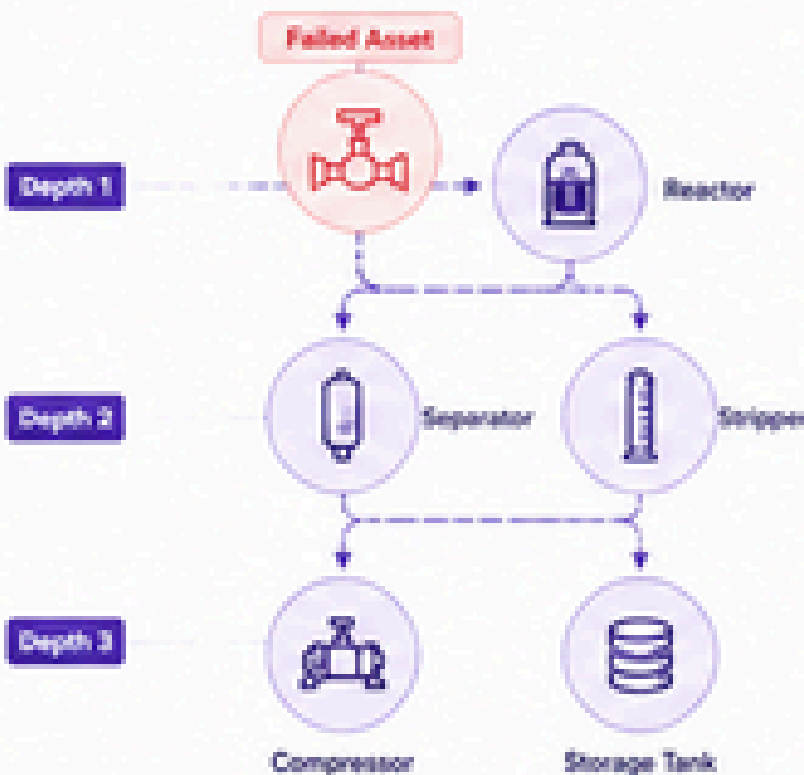
Every data point anchored to a Canonical Asset



AI traverses the Graph from this asset to deliver precise, context-aware RCA.

2 TOPOLOGICAL IMPACT ANALYSIS

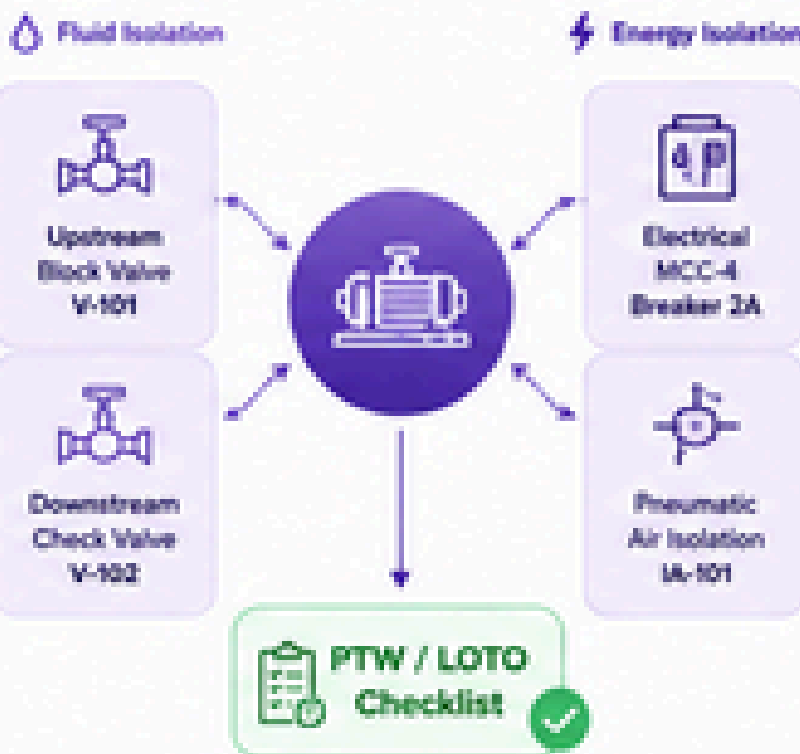
Calculate the "Blast Radius" of a Failure



Graph traversal maps downstream impacts so operators act before the cascade occurs.

3 AUTONOMOUS PTW & LOTO

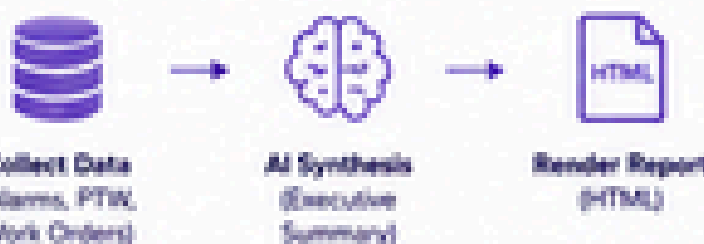
AI drafts safe isolation plan in seconds



Bi-directional graph walk finds exact isolation points. Technician gets a ready-to-execute checklist.

4 AUTOMATED REPORT GENERATION

Shift handover, automatically created & distributed



SHIFT HANDOVER REPORT

Executive Summary

Active PTW / LOTO

5 Active Permits

Critical Alarms

3 High-high 7 High 5 Low

Key Work Orders

4 In Progress



Beautiful HTML reports delivered via Email & Slack. No knowledge lost during shift changes.



Context-Aware AI
100% asset-grounded intelligence



Proactive Operations
Act upstream. Prevent cascades



Safer Maintenance
Accurate LOTO. Zero guesswork



Time & Cost Saved
Minutes, not hours. Every time.

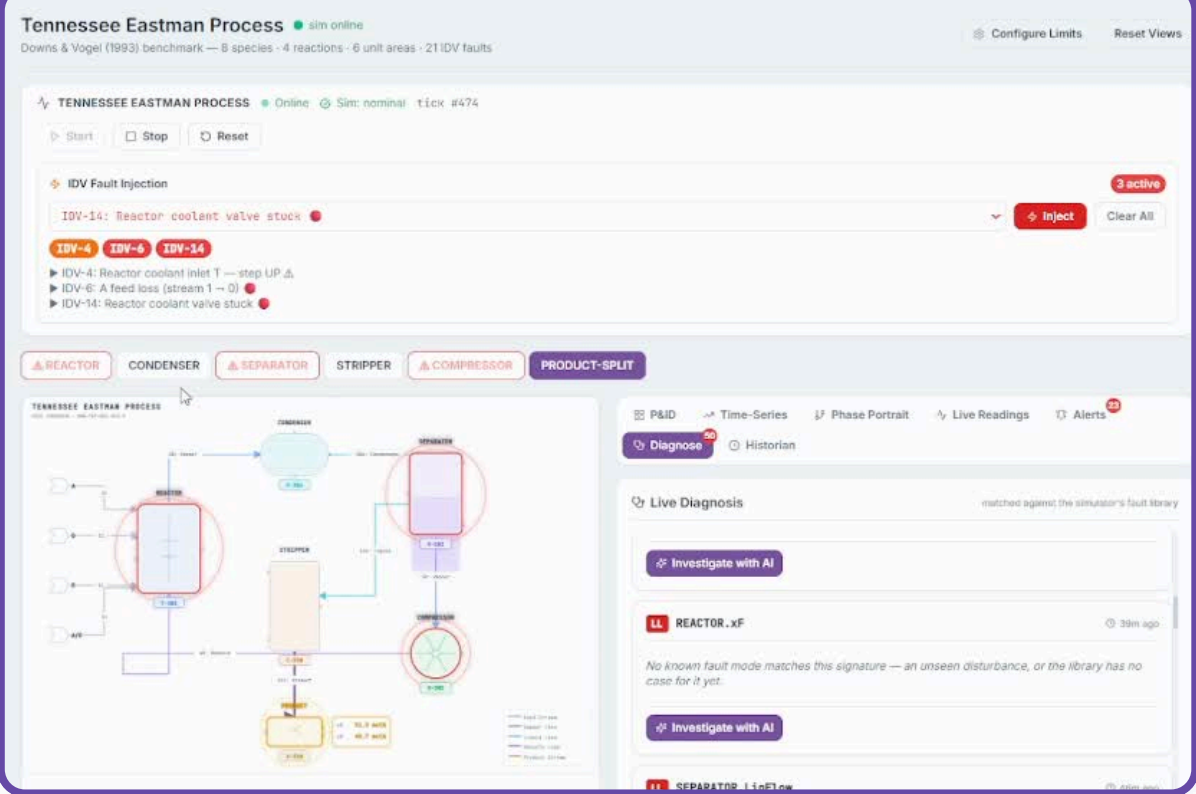


Knowledge Retention
Digitize expertise. Eliminate brain drain.



OFFICE COPILOT

Engineering & Control Room Command Center



Graph Exploration (PathRAG)

Deep Analytics & Trends

RCA & Work Order Review

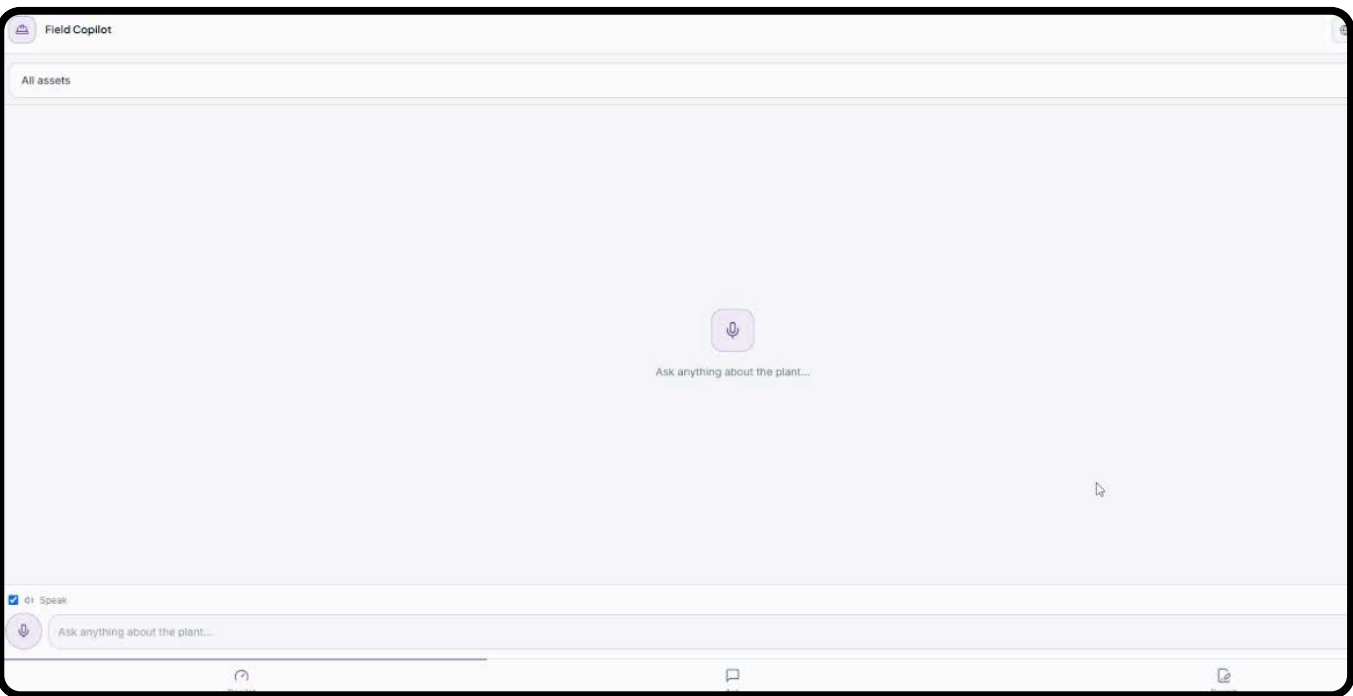
Slack & Gmail Integration

PLANTMIND
KNOWLEDGE GRAPH



FIELD COPILOT

Technician's Mobile Assistant



Context-Aware Work Queue

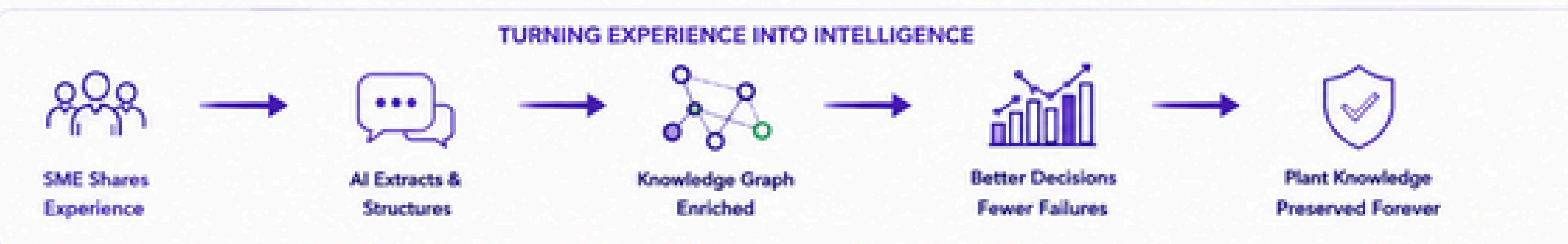
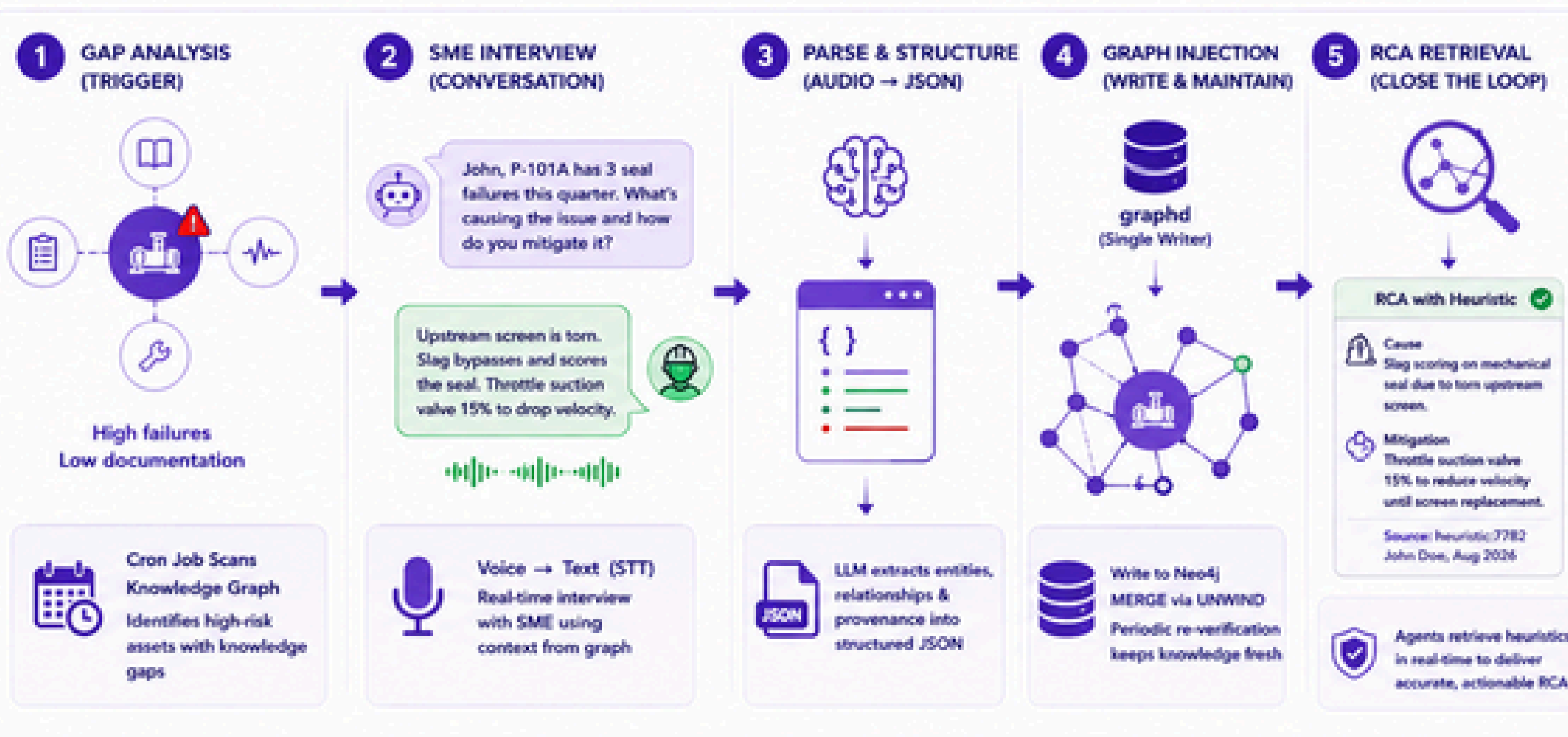
PTW / LOTO Checklist

Instant Answers (Oms Cache)

Capture & Share Field Knowledge

THE SEAMLESS HANDOFF – WORKFLOW IN ACTION





DEMO

[LIVE WALKTHROUGH](#)



PlantForge.ai

Unit 300

graph v24 - live

James Anderson
Process Engineer - Unit 300

PlantForge.ai

Interview in progress 00:01

MuteEnd Interview

Home

Ask

Alerts

Documents

Change Impact

Simulation

Fault Library

Asset Reports

Permits

Graph

Compliance

Interview

Profile

The interviewer is about to greet you...

COVERAGE · 0/0 TOPICS

ENGINEER

PlantForge.ai v2.0



Why PlantForge needs an MCP?



Engineers work where they are (IDE, Claude, local tools).



Secure, standard way to access plant data and tools.



Built-in guardrails keep access safe and governed.



Speeds up decisions with live + historical context.



Grounded answers with verifiable citations.



MCP CLIENT
Claude Desktop / IDE
`'claude mcp add'`

launches & owns the process



STDIO TRANSPORT · JSON-RPC 2.0
stdout = protocol ONLY
all logs → **stderr** (guarded)

tools/list
tools/call



TRANSPORT LAYER



stdio transport
JSON-RPC 2.0



stdout reserved for protocol



all logs → stderr (logging_guard)

BENEFITS



Use any MCP-compliant client (Claude / IDEs)



Security posture never bypassed



Low latency graph reads



Testable, reliable, maintainable

FastMCP("plantmind")

instructions: cite plant facts, never invent them

TOOLSET

TOOL(S)

→ BACKEND



QaToolset

ask_plant



GATEWAY

retrieve + cite



ChangeToolset

assess_change



GATEWAY

MoC impact



StatusToolset

plant_status



GATEWAY

/metrics (1 Hz)



GraphLookupToolset

get_failure_history



GRAPH (direct read)

get_connected_equipment



GRAPH (direct read)

get_work_orders · get_* (9 total)



GRAPH (direct read)

POLICY PATH

auth · rate-limit · RLS

READ PATH

least privilege



GATEWAY (HTTP)

POST /ask (QA)

POST /moc/assess (MoC)

GET /metrics (telemetry)

→ the ONE policy boundary



GraphBackend

AgentReader → Neo4j

READ-ONLY · lazy connect (fails per-call, NOT at launch)



Neo4j Graph (Knowledge Graph)



TimescaleDB (Time-Series)



MinIO Object Storage (Documents, Images, Artifacts)



DESIGN PRINCIPLES



Policy-bearing ops go THROUGH gateway (auth · rate-limit · RLS)



Pure structured reads hit Neo4j directly via READ-ONLY user



Backends are injectable → unit-test everything without gateway / Neo4j



Fail-safe startup lazy connections graceful per-call failures



DUAL-PATH STRATEGY



Through Gateway
High risk / high policy Checks & heavy ops

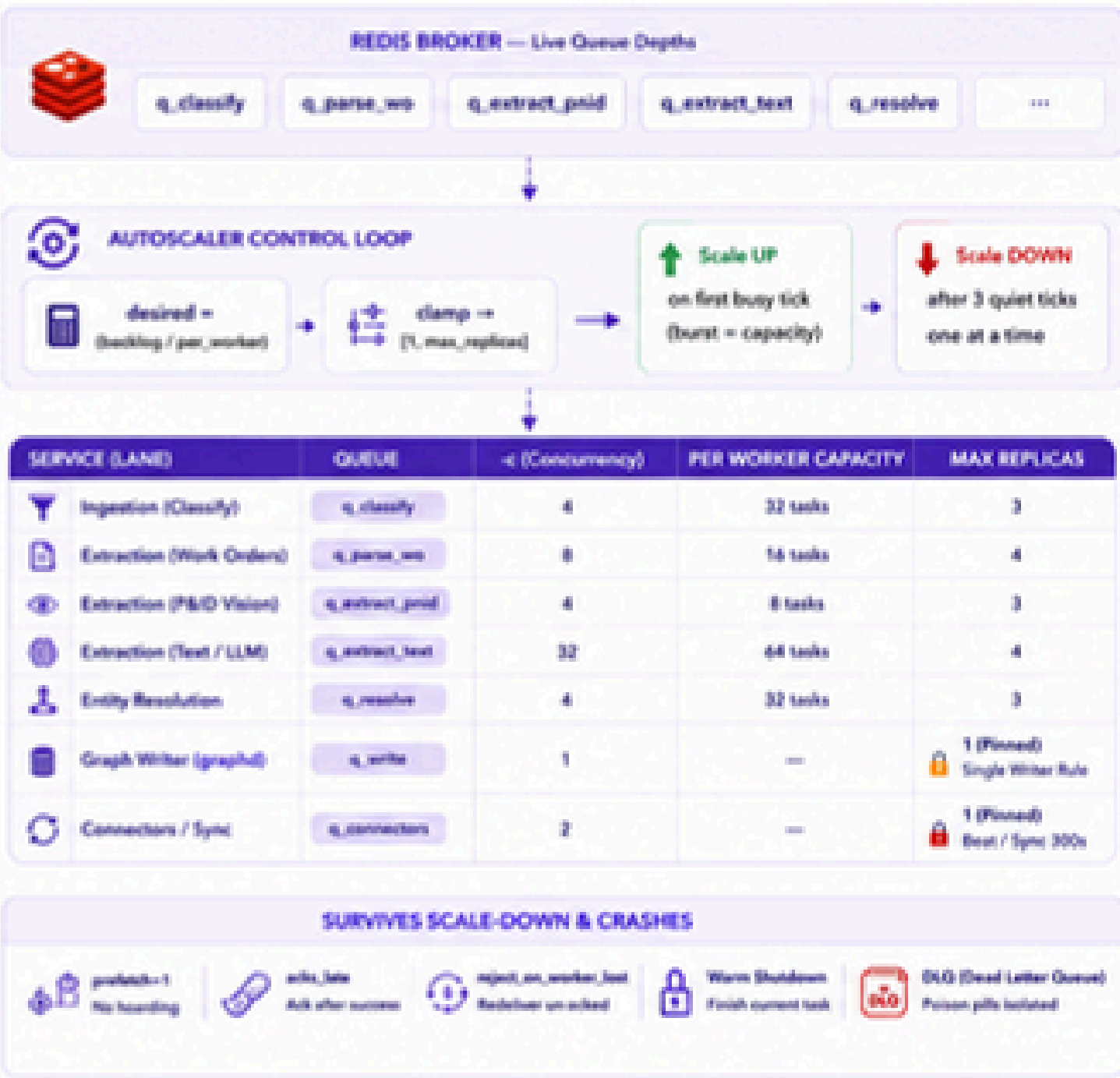


Direct to Graph
High frequency reads Fast & efficient



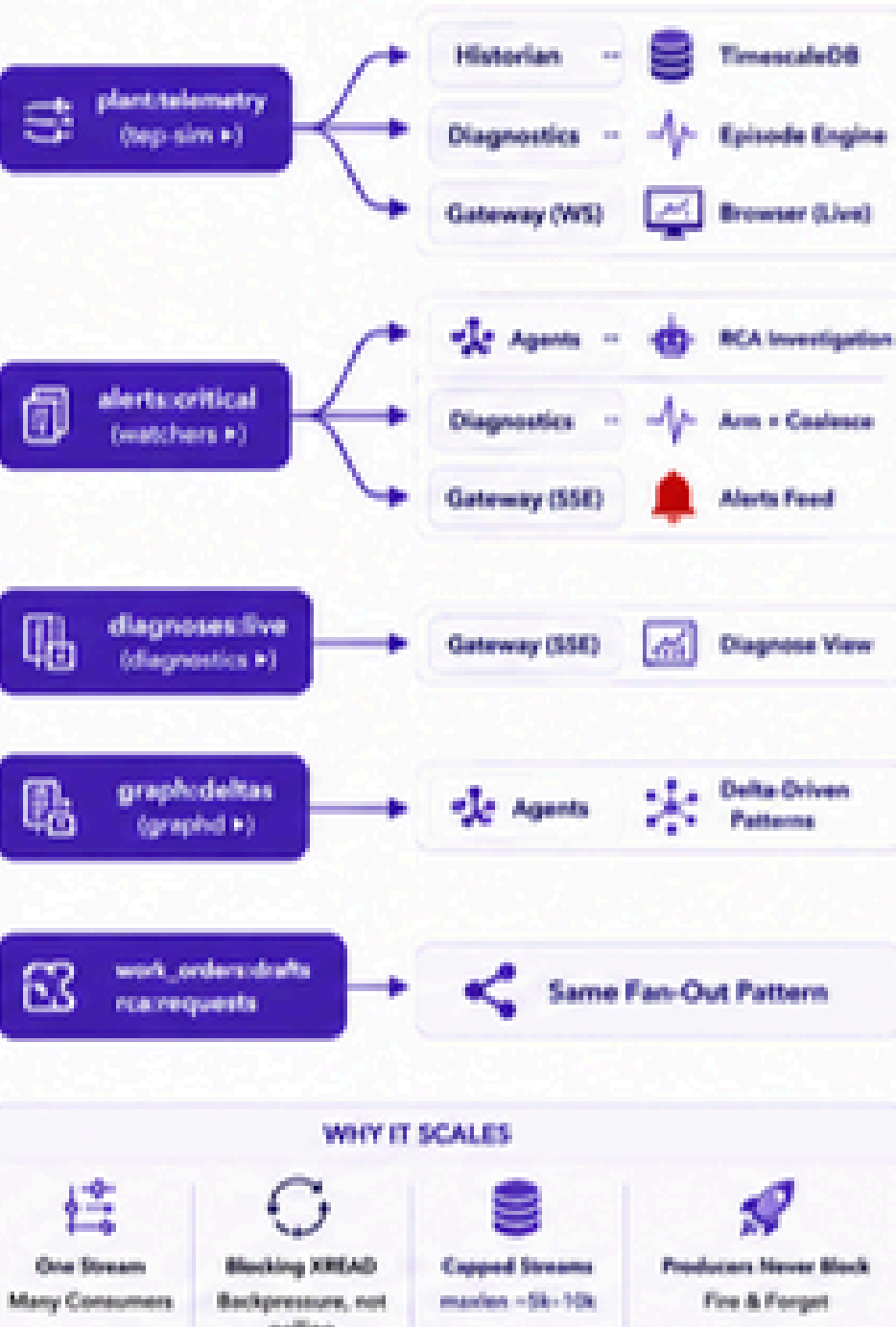
1 ELASTIC WORKER FLEET

Queue-Depth Autoscaling (poll every 5s)



2 REAL-TIME STREAMING BACKBONE

Redis Streams • Fan-Out • At-Least-Once



3 READ / ANSWER PATH

Concurrency Governors



FOUNDATION GUARANTEES



Idempotency
content-hash + MERGE + prev_hash



Exactly-Once Effect (Practically)
At least once delivery + idempotent ops



Observability
Metrics + Traces + Queue Depths



Resilience
DLQ + Retries + Isolation

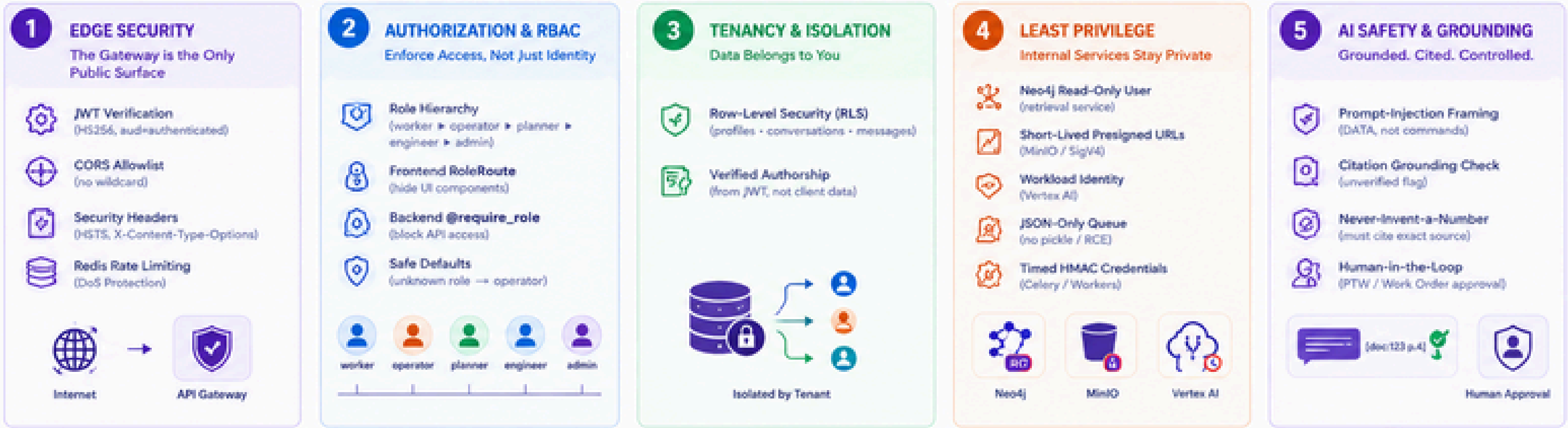


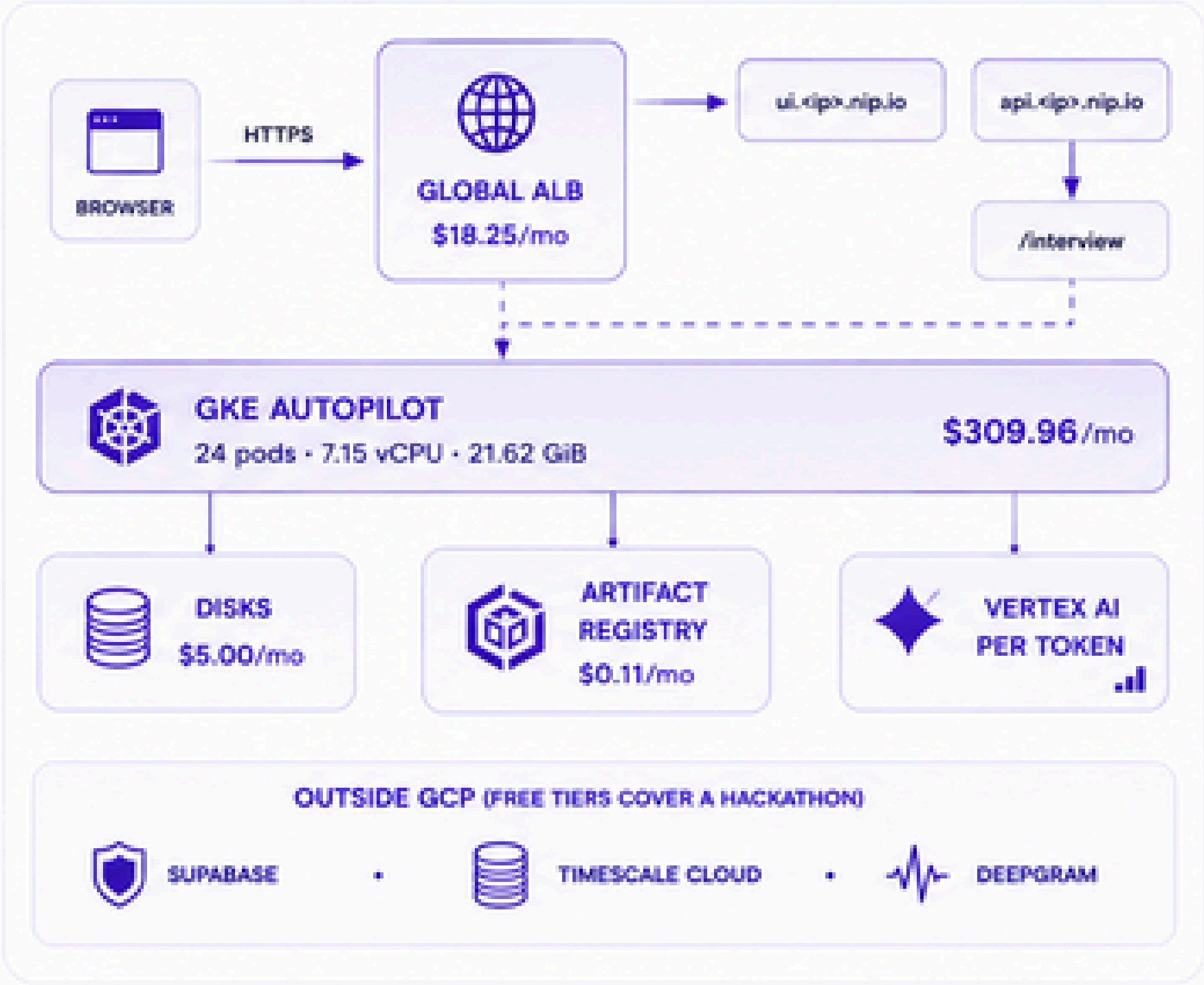
Cost Efficient
Scale to Zero ready lanes



RESULT

High Throughput • Low Latency
Resilient • Cost Optimized





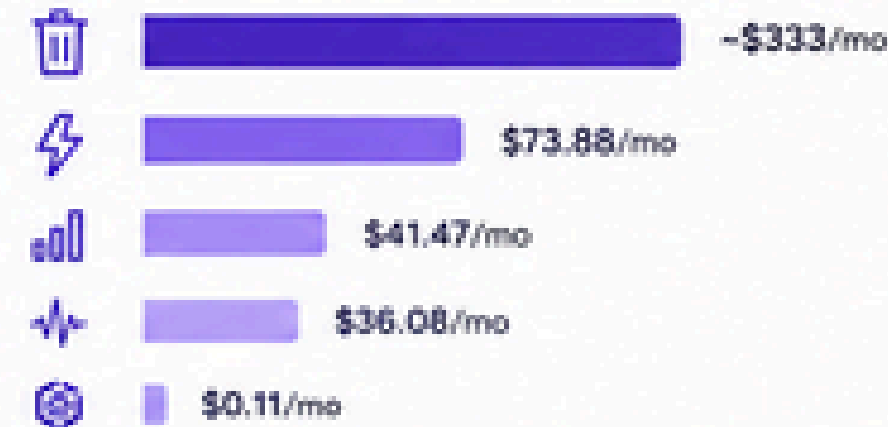
MONTHLY RUN-RATE



Autopilot vCPU	\$232.27	70%
Autopilot Memory	\$77.69	23%
Load Balancer	\$18.25	5%
Persistent Disks	\$5.00	2%
Artifact Registry	\$0.11	<1%

INFRA TOTAL \$333.32 /mo
≈ \$10.96/day | ≈ \$0.46/hour

BIGGEST SAVINGS



ONE-OFF INGESTION



DEMO-DAY POSTURE
~\$223/mo ≈ \$7.35/day

JUDGING WINDOW (8H)
~\$3.70 for the day

NOT
BILLED

TLS CERT
FREE

IP
STATIC IP
FREE

FIREWALL RULE
FREE

CLUSTER MGMT FEE
FREE (1st cluster)

TURN RELAY
NOT DEPLOYED



Thank You



Appendix



ADVANCED P&ID DIGITIZATION and Topological Graph Ingestion



Goal: Convert dense, high-resolution P&IDs into an accurate, fully routed topological Knowledge Graph without losing microscopic details.

THE CHALLENGE



Industrial P&IDs are dense, high-resolution schematics where critical details (arrows, reducer sizes, valve states) are microscopic.



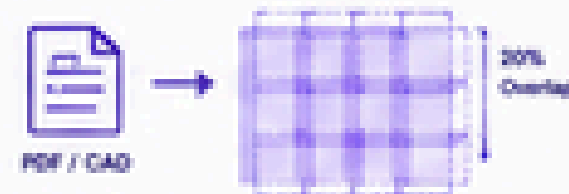
Standard VLMs downsample images, leading to hallucinated tags and broken topological paths.



PlantForge solves this with a hybrid deterministic-semantic pipeline.

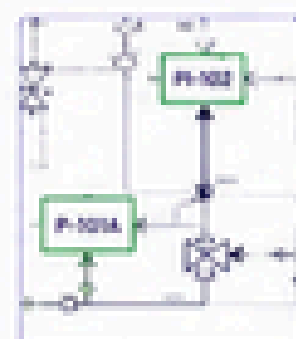
A7.1 THE P&ID DIGITIZATION PIPELINE

1 Pre-Processing (OpenCV Pyramidal Tiles)



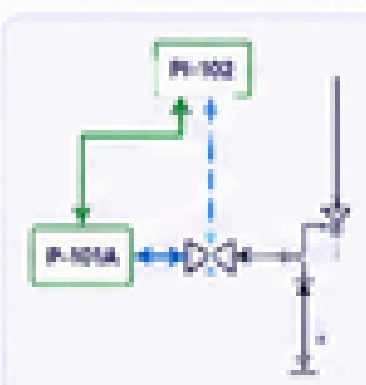
- ✓ High-DPI rasterization (≥ 600 DPI) into a baseline matrix
- ✓ Overlapping slices (e.g., 1024x1024 px) preserve edge continuity

2 Tag Extraction (Deterministic OCR)



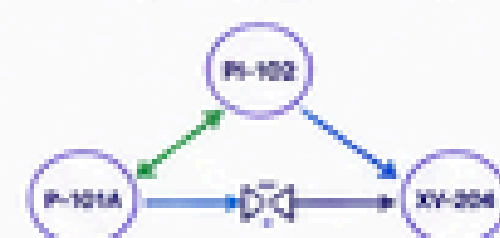
- ② OCR extracts text + bounding boxes (x0, y0, x1, y1)
- ③ ISA tag grammar filtering
- ④ Spatial index of validated tags

3 Path Tracing (VLM Subgraph Routing)



- ⑤ VLM receives tile image + tag index
- Traces lines, detects flow direction (forward, bidirectional, unknown)

4 KG Assembly (Global Coordinate Map)



- ✓ Stitch subgraphs using overlaps & global coordinates
- ✓ Final Knowledge Graph emitted with bounding boxes

A7.2 PHASE 1: OPENCV PYRAMIDAL TILING



- 1 High-DPI Rasterization
Convert vector PDF/CAD to raster at ≥ 600 DPI.



- 2 Overlapping Slices
Slice into high-res tiles (e.g., 1024x1024 px) with 20% spatial overlap.



- 3 Edge Preservation
Overlap prevents lines from being cut at tile boundaries.

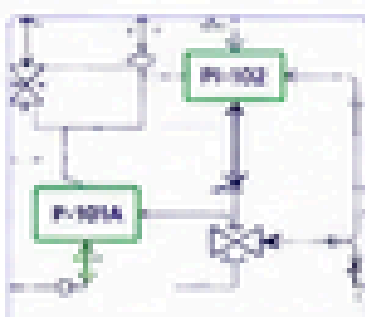


Preserves microscopic details like arrowheads, reducer sizes, and valve states for accurate downstream analysis.

A7.3 PHASE 2: DETERMINISTIC OCR & TAG LOCALIZATION



Build a deterministic ground truth of all text entities.



- ✓ OCR (or vector extraction) returns text + bounding boxes (x0, y0, x1, y1)
- ✓ ISA tag grammar validation (e.g., PI-102, P-101A)
- ✓ Spatial index of validated tags with coordinates

Example Tag Index

Tag	Text	Type	Bounding Box (x0, y0, x1, y1)
T1	P-101A	EQUIPMENT	[150.2, 450.0, 180.5, 465.1]
T2	PI-102	INSTRUMENT	[120.1, 220.3, 145.7, 235.0]
T3	XV-204	VALVE	[800.6, 450.8, 850.2, 465.6]
...	...	...	...



Deterministic localization prevents hallucinated tags and misreads.

A7.4 PHASE 3: VLM SUBGRAPH ROUTING



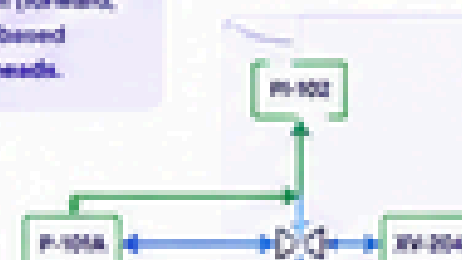
Trace connections between known tag anchors.

PASS-2 VLM PROMPT

These components and their bounding boxes were identified in this section: {file_tag_list}

Trace the process lines connecting these components. List every direct connection between two tags. Determine flow direction (forward, bidirectional, unknown) based strictly on visible arrowheads.

- Forward
- ↔ Bidirectional
- Unknown



High-res tiles + deterministic anchors = accurate line tracing and flow direction.

A7.5 PHASE 4: PATH KNOWLEDGE GRAPH (KG)



Coordinate Stitching
Overlaps + global coordinates merge subgraphs into a continuous topology.



Candidate Emission
Emit CandidateSubgraph payload for the resolver.

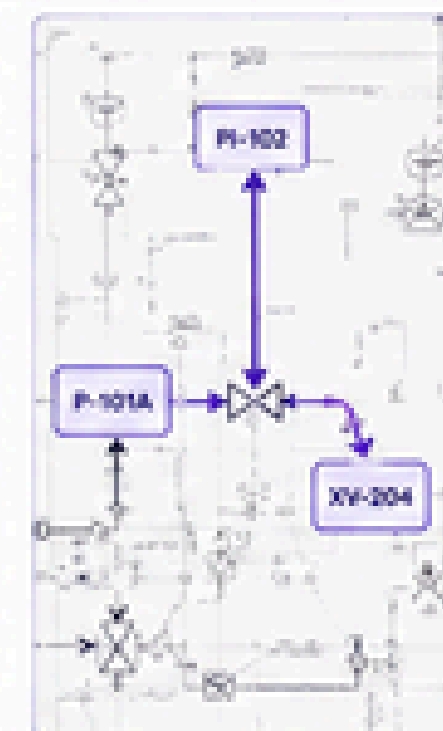
```
CandidateNode(  
  type=EQUIPMENT,  
  surface_name="P-101A",  
  props={"bbox": [150.2, 450.0, 180.5, 465.1]}  
)  
...  
CandidateEdge(  
  type=CONNECTED_TO, src="P-101A",  
  dst="PI-102", props={"direction": "forward"}  
)
```

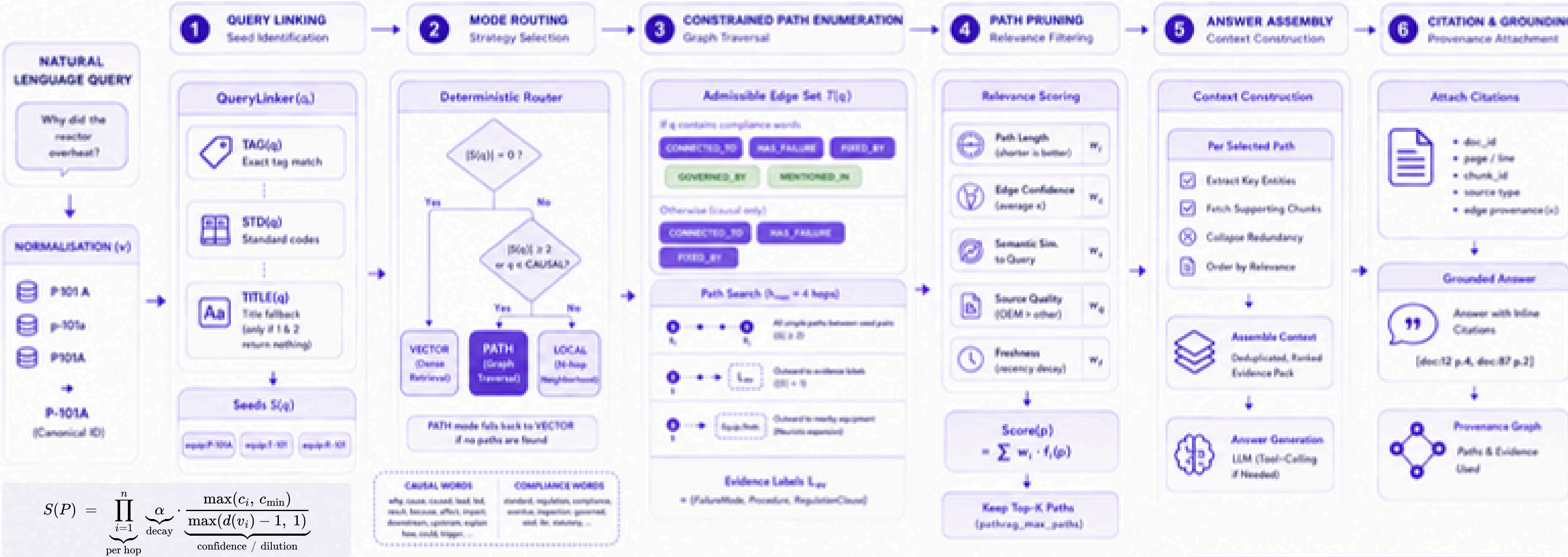


Visual Graph Sync
UI overlays original P&ID with extracted graph for click-to-highlight verification.



True topological map enables accurate RCA, isolation boundary walking, and traceability.







SIMULATION STACK (All run at 1 Hz)



runner.py
Wall-clock loop
1 Hz tick



controller.py
PID bank
→ MV vector $u(t)$



model.py
solve_ivp (RK45)
 $y(t) \rightarrow y(t+dt)$



publisher.py
Tags → Redis Stream
→ Historian



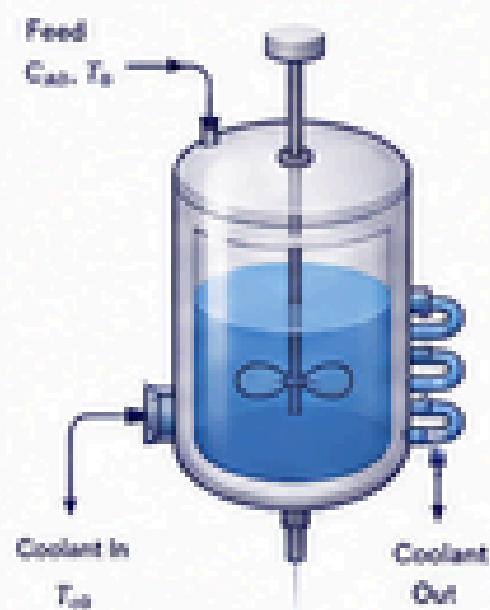
TimescaleDB
Hypertable



Alarm Watcher
ISA-18.2
Compliance

1 CSTR – Exothermic Reactor

3 states: $[C_A, T, T_c]$



Governing Equations

$$\frac{dC_A}{dt} = \frac{C_{A0} - C_A}{\tau} - k(T) C_A$$

$$\frac{dT}{dt} = \frac{T_0 - T}{\tau} + \frac{(-\Delta H_r)}{\rho C_p} k(T) C_A - \frac{UA}{V \rho C_p} (T - T_c)$$

$$\frac{dT_c}{dt} = \frac{F_c}{V_c} (T_{c0} - T_c) + \frac{UA}{V_c \rho_c C_{pc}} (T - T_c)$$

MV (Cooling Valve)

$$F_c = \frac{u}{100} F_{c,max}$$

u = % open

Kinetics (Arrhenius)

$$k(T) = k_0 \exp\left(-\frac{E_a}{RT}\right)$$

Key Parameters

k_0	$7.2 \times 10^{10} \text{ s}^{-1}$
E_a	$72,750 \text{ J/mol}$
ΔH_r	$-5.0 \times 10^4 \text{ J/mol}$
R	$8.314 \text{ J/mol}\cdot\text{K}$
V	1.0 m^3
Q_{feed}	$0.1 \text{ m}^3/\text{s}$

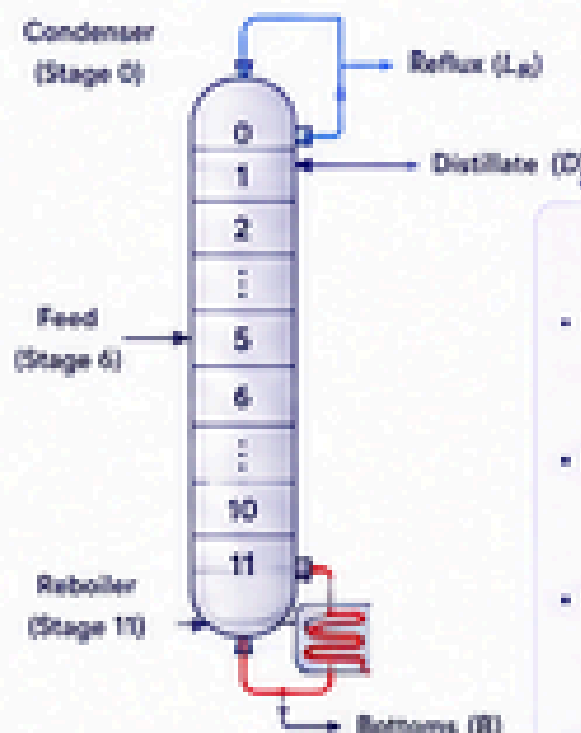
UA	$5,000 \text{ W/K}$
ρC_p	$5.0 \times 10^3 \text{ J/m}^3\cdot\text{K}$
$\rho_c C_{pc}$	$4.2 \times 10^3 \text{ J/m}^3\cdot\text{K}$
V_c	0.14 m^3
C_{A0}	4.0 mol/L
$F_{c,max}$	$0.05 \text{ m}^3/\text{s}$



Reaction heat generation rises exponentially with T while heat removal is linear → possible thermal runaway.

2 Binary Distillation Column

12 stages (0 = condenser, 6 = feed, 11 = reboiler)



VLE ($\alpha = 2.5$)

$$y_i = \frac{\alpha x_i}{1 + (\alpha - 1)x_i}$$

Stage Balance (examples)

- Rectifying ($1 \leq i \leq 5$)

$$\frac{dx_i}{dt} = \frac{L_R(x_{i-1} - x_i) + V(y_{i+1} - y_i)}{M_{tray}}$$

- Feed Stage ($i = 6$)

$$\frac{dx_6}{dt} = \frac{L_R x_5 - L_F x_6 + V(y_7 - y_6) + F x_F}{M_{tray}}$$

- Stripping ($7 \leq i \leq 10$)

$$\frac{dx_i}{dt} = \frac{L_R(x_{i-1} - x_i) + V(y_{i+1} - y_i)}{M_{tray}}$$

- Reboiler ($i = 11$)

$$\frac{dx_{11}}{dt} = \frac{L_R(x_{10} - x_{11}) - V(y_{11} - x_{11})}{M_{reb}}$$

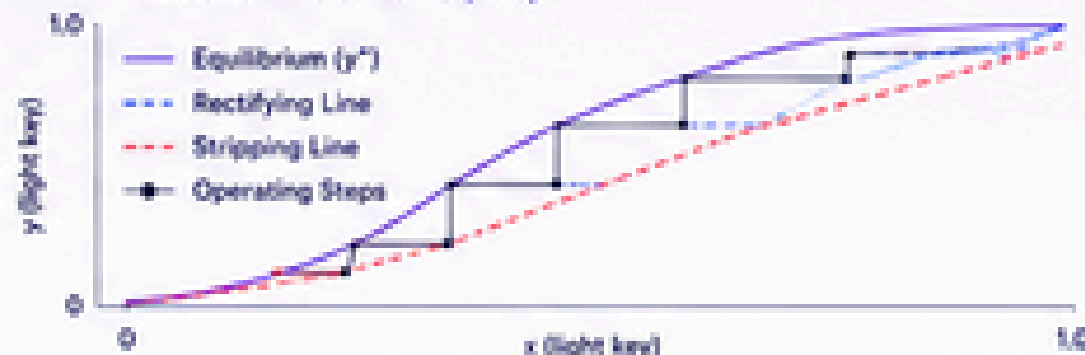
Manipulated Variables

- Reflux Flow L_R
- Boil-up Flow V (via reboiler duty)

Holdups

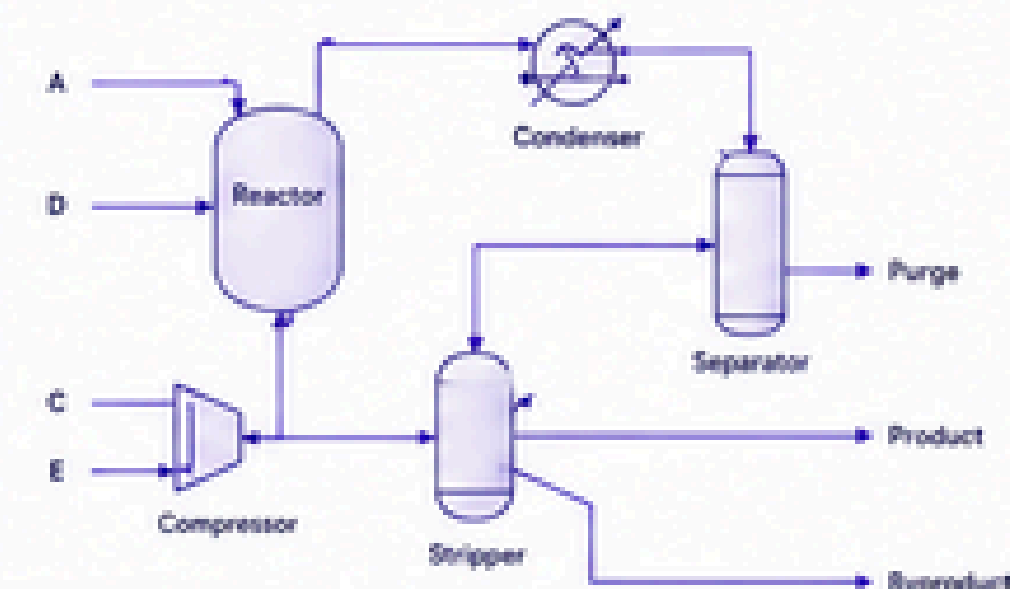
- $M_{tray} = 0.5$
- $M_{tray} = 0.1$
- $M_{reb} = 0.5$
- $M_{reb} = 1.0$

McCabe-Thiele Profile (Live)



3 Tennessee Eastman Process (TEP)

Nonlinear process benchmark



State-Space Model

$$\frac{dx}{dt} = f(x, u, d)$$

x : states (41)

u : manipulated vars (12)

d : disturbances (20)

y : measured outputs (41)

Fault Injection (IDVs)



Valve Stiction



Sensor Bias



Heat Exchanger Fouling



Compressor Efficiency



Reaction Kinetics

Outputs



41 Measurements



1 Hz Telemetry to Redis



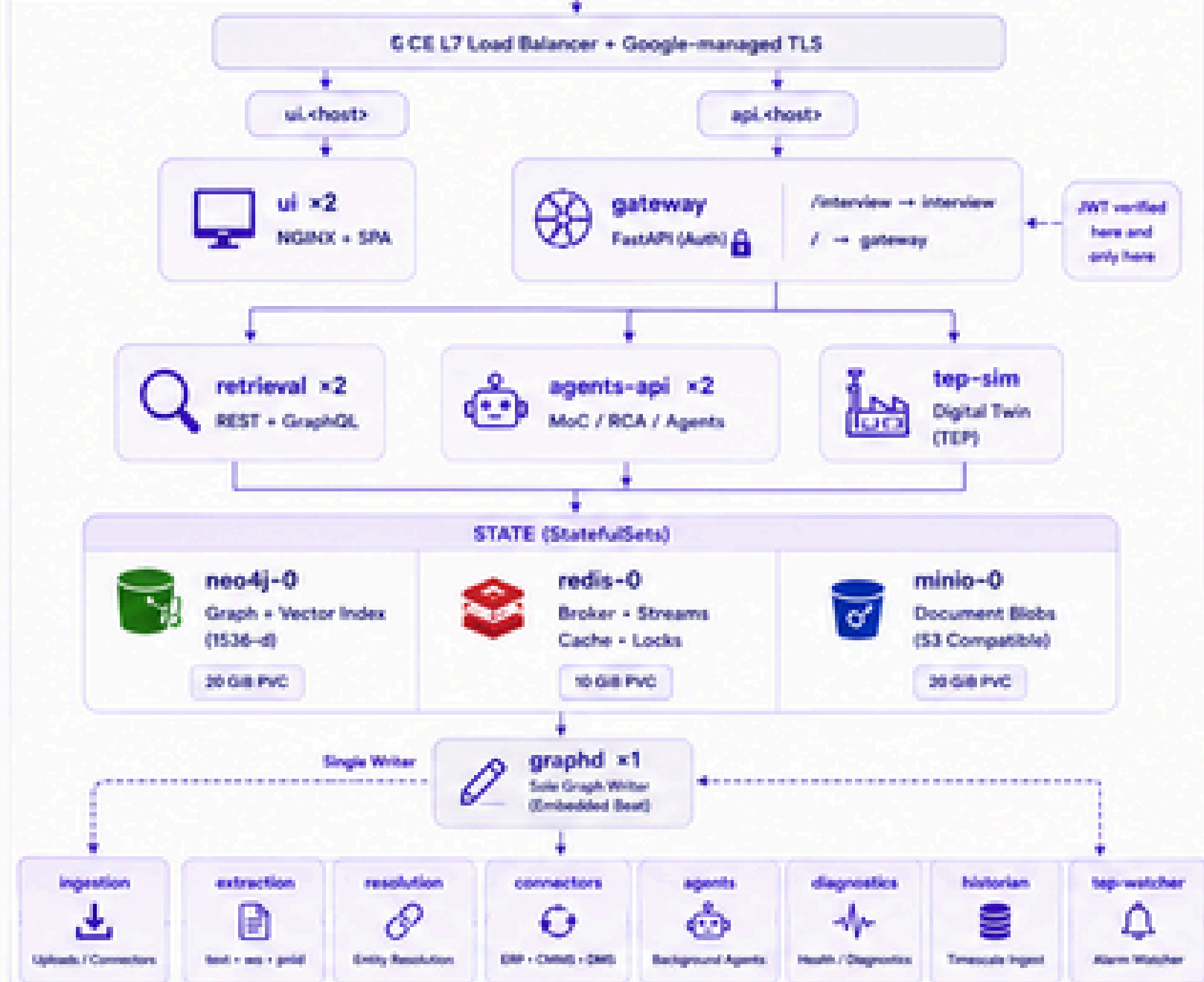
Events to Watchers



A4.1 SERVICE TOPOLOGY



Internet



A4.3 AS CELERY PIPELINE (HIGH LEVEL)



A4.2 REDIS - 5 DISTINCT ROLES



A4 CELERY DELIVERY POLICY

task_serializer	json
accept_content	["json"]
task_ignore_result	True
task_ackslate	True
task_reject_on_worker_lost	True
worker_prefetch_multiplier	1
task_soft_time_limit	540s
task_time_limit	600s

★ At-least-once delivery
Idempotent by design
(hash gate + MERGE + prov_hash)

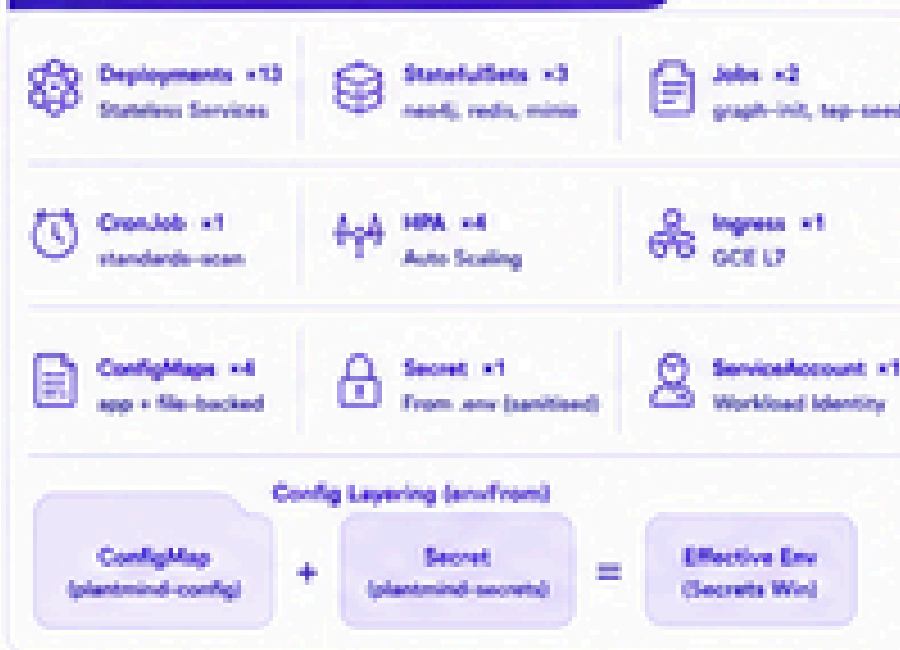
A5 GATEWAY (REST ENDPOINTS)

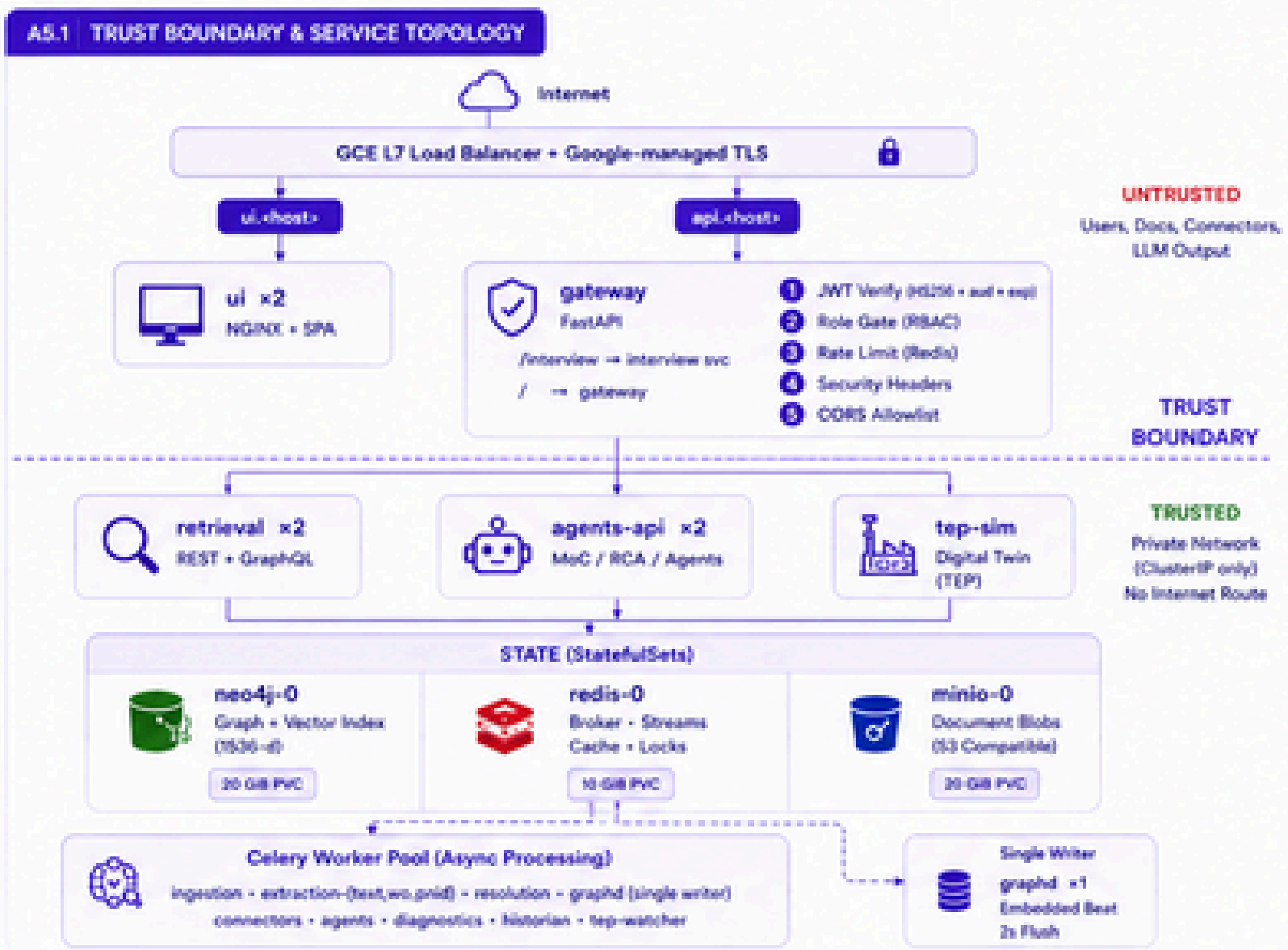
• POST /ask	• POST /mcu/assess
• POST /ask/stream	• GET /graph/snapshot
• POST /documents/upload	• GET /work-orders
• GET /documents/{id}	• GET /compliance
• GET /events/alerts (SSE)	• POST /permit
	• GET /health (open)

A4.7 DATA STORES

Neo4j 5	Graph + Vector Index Unified traversal & semantic search
Redis 7	Broker + Streams + Cache + Locks Event log & coordination
MinIO	Raw Document Blobs S3 compatible (presigned URLs)
TimescaleDB	Time-series Historian Simulation data
Supabase	Auth + Profiles + Conversations Postgres + RLS (JWT issuer)

A4.4 KUBERNETES (GKE AUTOPILOT)





A5.2 THREATS ADDRESSED & CONTROLS

1	Unauthenticated Access	→	JWT verify (HS256, aud, exp)	auth.py
2	Privilege Escalation	→	5-tier RBAC, per-router gating	deps.py
3	Cross-Tenant Data Access	→	Postgres RLS (tenant isolation)	schema.sql
4	RCE via Task Deserialization	→	JSON-only serializer	worker_app.py
5	Stored XSS / Content Sniffing	→	nosniff, CSP, blob: rendering	security.py
6	Clickjacking	→	X-Frame-Options: DENY	security.py
7	CORS Misconfiguration	→	Allowlist, no credentials	main.py
8	Abuse / Cost Exhaustion	→	Redis rate limit on subject (sub)	ratelimit.py
9	Credential Exposure	→	Workload Identity (no keys)	50-sa.yaml
10	TURN Password Leak	→	HMAC time-limited credentials	turn.py
11	Fabricated Citations (LLM)	→	Deterministic grounding check	grounding.py
12	Ingestion Integrity	→	Hash gate + release on fail	service.py
13	Graph Corruption	→	Single writer + Redis mutex	graphd.py

A5.3 CONTROL MAP

Threat	Control	Where	Threat	Control	Where
Forged / replayed token	HS256 + aud + exp	auth.py	Pickle RCE	json-only serializer	worker_app.py
Algorithm confusion	algorithms pinned	auth.py	Stored XSS / sniffing	nosniff, CSP, blob: URL	security.py
Privilege escalation	role hierarchy, safe default	deps.py	Clickjacking	X-Frame-Options: DENY	security.py
Endpoint added unprotected	per-router dependency	main.py	CORS over-permission	allowlist, no credentials	main.py
Cross-tenant read	Postgres RLS	schema.sql	Abuse / cost exhaustion	Redis fixed-window on sub	ratelimit.py
RLS policy recursion	JWT claim, not table read	schema.sql	SA key leakage	Workload Identity	50-sa.yaml
Search-path hijack	security definer + pinned path	schema.sql	TURN password leak	HMAC time-limited cred	turn.py
			Fabricated citation	deterministic grounding	grounding.py
			Duplicate / partial ingest	hash gate + release	service.py
			Graph corruption	single writer + mutex	graphd.py

A5.4 KNOWN GAPS — STATED, NOT HIDDEN

1 Auth Can Be Disabled
If SUPABASE_JWT_SECRET is unset, gateway runs open and grants engineer. Logs announce: "gateway is OPEN". Must be set in real deployments.

2 Wide UDP Ingress for WebRTC
udp:32768-60999 open to 0.0.0.0 for alerts. Tight alternative: TURN relay on fixed ports. Known, reversible exposure.

3 Prompt Injection via Documents
Malicious instructions in ingested documents could influence LLMs. Partially mitigated (schema + grounding). No dedicated injection filter.

4 No Audit Log (Yet)
Access and actions are not recorded in an immutable audit trail. Planned for production deployments.

\$

Billed on the sum of your pods' resource **REQUESTS**, per second.



 An over-requested pod costs exactly as much idle as it does busy. Right-sizing requests is the cost lever.

7.15 vCPU
21.63 GiB Memory

- 22 Workloads (steady state)
- 7.15 vCPU
- 21.63 GiB
- Total Requests

Workload	vCPU	Memory	Share (vCPU-Mem)
extraction-text	1.00	4 GiB	~24%
top-sim	1.00	1 GiB	~18%
retrieval x2	1.00	2 GiB	~17%
extraction-res	0.50	3 GiB	~12%
Everything else	3.65	11 GiB	~31%
TOTAL	7.15	21.63 GiB	100%

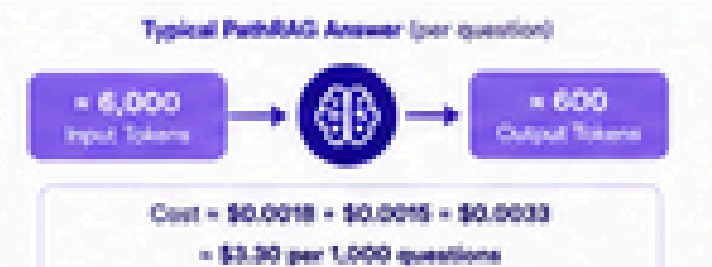
Storage (PVCs)
neo-ij: 20 GiB • minio: 20 GiB • redis: 10 GiB
Total: 50 GiB

Network & Other
1 x L7 Load Balancer • 1 x Static IP • 1 x Managed Cert
2 hostnames (nl, api) • 22 images in Artifact Registry

Component	Quantity	Rate (Indic.)	Monthly (USD)
Compute (Autopilot)			
vCPU	215 vCPU	\$0.0573 / vCPU-hr	~\$299
Memory	21.63 GiB	\$0.00633 / GiB-hr	~\$100
Compute Subtotal			~\$399
Platform & Operations			
Cluster management fee	1 cluster	\$0.10 / hr	\$73
Less free tier credit	-	~\$74.40	~\$0
Storage & Networking			
PD - Balanced	50 GiB	\$0.10 / GiB-mo	~\$5
L7 Load Balancer	1 rule	\$0.025 / hr	~\$18
Static IP	1 (in use)	Free while attached	\$0
Artifact Registry	~10 GiB	\$0.10 / GiB-mo	~\$1
Egress	Variable	\$0.12 / GiB	~\$1-10

TOTAL INFRASTRUCTURE = \$425 / month

Model Tiers		
	CHEAP Gemini 1.5 Flash-Lite	Classification, condensing, eval judge
	MID Gemini 1.5 Flash	Extraction, answer generation
	VISION Gemini 1.5 Flash	RAG, nameplates, charts, OCR



Per Document Ingestion (one pass)			
 Short Text/ Email	 Table (Block Order)	 Web/Image (Mixed)	 PDF page Manual
~\$0.005	~\$0.003	~\$0.02	~\$0.05 - 0.1

- **Answer Cache**
Recent questions = 50
- **Extraction Cache + Lock**
Re-index identical index = 50
- **Content-Hash Gate**
Duplicate never enter
- **Rate Limiting**
Crawls worth data source

	HACKATHON 3-day demo	PILOT 1 plant	PRODUCTION Multi-unit
Cluster Hours	72 h	720 h	720 h
Questions / Day	~50	~500	~5,000
Documents Ingested	44	~2,000	~50,000
Infrastructure	~\$42	~\$425	~\$900*
LLM Inference	~\$2	~\$60	~\$1000
TOTAL	~\$44	~\$485	~\$1,450
Cost per Question	—	~\$0.032	~\$0.010

 Marginal cost per question **FALLS** with scale.
Infrastructure is fixed, cache hit rate rises.

Item	Impact	Effort	Priority
1 Delete the cluster from demos gitpod container clusters delete planimol --region us-central1	HIGHEST IMPACT	Low	High
2 Scale non-demo workloads to zero top-sim, diagnostics, historians, top-watcher, connectors	HIGH	Low	High
3 Right-size extraction-test between bursts Scale to 0 when not ingesting	MEDIUM	Low	High
4 Drop HA pairs in non-production retrieval, gateway, agents-api, ui (x2 → x1)	LOW-MED	Low	High
5 Already done — keep it - FastEmbed local - Answer cache - Extraction cache - Flash-Lite - Rate limiting	LOW (ONGOING)	Low	High

Report Audit, profiles, communications	Timescale Cloud Historical Replication	Diagram Visual DDL/DML	Domain Replicates (1M)
Free tier up to \$25/mo	Free tier up to \$50/mo	Free per minute	\$0

 Need a Native Vector Index No separate vector DB	 Facilitated In Progress \$0 embedding spend (embeds only, RAGs)	 Ready Streamline Not Kafka No extra cluster	 Rule-Based Router No LLM call to decide strategy	 Democratize Grounding No second LLM call to grade provenance	 Autopilot Billed on requests, not nodes	 Planned LLM Routing Flash-Lite for cheap, the Flash for heavy work
---	--	--	---	---	--	---

```

1- # CPU / Memory requests (what Autoscaled sets)
2- kubectl -n plantoid get pods \
3-   --output-columns=NAME,status,cpu,CPU_request,memory,memory_request,READY,memory_request,READY
4-
5- # Storage
6- kubectl -n plantoid get pv
7-
8- # Live utilization (to find over-requested pods)
9- kubectl -n plantoid top pods

```



Our system is highly **adaptable** and **versatile**, making it ideal for a range of applications that helps ass an unified **asset management** and **intelligence platform** for Process Engineering Plants, including everything from advanced **question answering** to the generation of **high-quality reports**, along with other essential functions.PlantForge translates **complex plant topology** into frontline action—streamlining regulatory compliance, accelerating root-cause analysis, and enforcing process safety before a single wrench turns.

